

Comparing Homeownership in Germany and Italy: Analyzing the Role of Relevant Policies through the Lense of a Quantitative Model

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1 Introduction

Homeownership rates differ widely across developed countries. Around the 2010's, about 44% of German households owned their main residence, compared to 67% in the United States and around 69% in Italy.¹ Understanding these persistent differences matters because they shape, among other things, individual precautionary saving behavior (Cocco 2005; Fernández-Villaverde and Krueger 2011) and mobility in response to labor market opportunities (Oswald 1996). At the aggregate level, they also translate into differences in wealth inequality (Kindermann and Kohls 2018) and in the exposure of household balance sheets to monetary policy shocks (Auclert 2019). A large body of research has therefore started exploring, also quantitatively, the determinants of the choice between owning and renting and whether cross-country differences can be attributed to these determinants. A notable example is Kaas et al. (2021), who show that two thirds of the homeownership gap between Germany and the United States can be traced to three policies concerning housing markets: differences in real-estate transfer tax (RETT) magnitudes, the presence of a social housing program, and the deductibility of mortgage interest. This thesis takes their framework as its starting point and asks the analogous question for the Germany–Italy pair: can the homeownership gap between these two countries be explained by differences in housing policies, or do other factors play a decisive role?

Table 1 documents the homeownership gap among the two countries along the age and wealth dimensions. Italian values are computed from the Survey on Household Income and Wealth (SHIW) (Banca d'Italia 2025), pooled across the 2010 and 2014 waves; German values are taken from the German Socioeconomic Panel (SOEP) through Kaas et al. (2021). Along the age distribution, Italian households exhibit persistently higher homeownership rates than their German counterparts across all age groups, with a gap ranging from approximately 20 percentage points for young households (25–34) to over 30 percentage points for retirees (65+). The differences are also particularly pronounced between the third and seventh deciles.

Table 1: Homeownership rate in Italy and Germany: age and wealth distributions

<i>Panel A: By age group</i>										
Age	25–34	35–44	45–54	55–64	65+					
Italy (%)	40.0	57.3	68.7	77.5	77.2					
Germany (%)	17.1	40.9	48.2	54.0	46.6					

<i>Panel B: By wealth decile (working-age households)</i>										
Decile	1	2	3	4	5	6	7	8	9	10
Italy (%)	1.4	2.2	12.0	59.9	88.6	93.1	95.5	97.6	96.1	97.5
Germany (%)	9.8	0.6	2.7	6.5	13.0	41.2	69.2	87.3	93.0	94.5

Notes. Panel A reports homeownership rates by age bin for households whose head is older than 24; Panel B reports homeownership rates by working-age wealth decile for households whose head is between 25 and 64. Italian values are computed by the author from the Survey on Household Income and Wealth (SHIW) of Banca d'Italia for the years 2010 and 2014 (see Appendix G); German values are taken from the estimation of Kaas et al. (2021) on the wealth modules of the German Socio-Economic Panel (SOEP) for the waves 2002, 2007 and 2012.

¹German values are obtained from the German Socio Economic Panel by Kaas et al. (2021), Italian values are computed by the author from the Survey on Household Income and Wealth data (see further in the text), US values are based on the US Census (2010) and presented by Kaas et al. (2021).

To test my hypothesis, I employ the overlapping-generations model with uninsurable income and housing risk and endogenous homeownership of Kaas et al. (2021), using their same calibration to the German economy. Among the institutional factors that differentiate housing in Italy from Germany, I conduct a series of counterfactual exercises by transplanting two distinctive features of the Italian housing system in the model: a reduced real-estate transfer tax, from 5.2% to 1.33%, and a different design of the social housing system. The Italian design enters the model through three features: a deeper rent discount relative to the market rate, a residualistic targeting of eligibility, and a right-to-buy (RTB) mechanism that allows tenants to purchase their unit at a statutory discount. I then analyze these policies first in isolation and then jointly, with the RTB mechanism also examined separately. To the best of my knowledge, this is the first implementation of a right-to-buy mechanism in a quantitative heterogeneous-agent model used to study the Italian social housing system.

Combining the RETT reduction with the Italian social housing design accounts for approximately 41% of the cross-country difference in homeownership. Taken independently, the RETT reduction accounts for almost one third of this gap, while implementing the Italian social housing design generates more modest effects, due to a crowding-out effect of RTB purchases on private homeownership. Interestingly, the Italian social housing design also alters the saving dynamics of social housing tenants, who accumulate additional financial wealth in anticipation of a future RTB offer, and transforms the role of the social housing sector into a safety net for the poorest households, with the RTB option providing a pathway to homeownership for tenants who would not otherwise have afforded a private-market purchase.

The welfare effects diverge across policies and across the income distribution. The RETT reduction lowers welfare, the most affected being the poorest households. The right-to-buy mechanism in isolation and the Italian social housing design produce positive welfare effects across the entire income distribution, with gains skewed toward lower-income households who benefit most from the policy change. Combining the RETT reduction and the Italian social housing design delivers welfare gains across all deciles.

The rest of the thesis proceeds as follows. Section 2 positions this thesis within the relevant branches of the literature. Section 3 describes the institutional framework that motivates the counterfactual exercises. Section 4 presents the benchmark model as in Kaas et al. (2021) and the calibration strategy. Section 5 introduces the counterfactual exercises and their implementation. Section 6 presents the homeownership results and discusses factors not captured in the model. Section 7 reports the welfare implications, and Section 8 concludes.

2 Related Literature

This work is at the intersection of three strands of literature: incomplete markets overlapping-generations models with housing, structural policy evaluation of housing instruments, and cross-country decompositions of homeownership differences. This section surveys these strands with emphasis on the features that Kaas et al. (2021) draw upon and on which this thesis builds.

2.1 Quantitative Models of Housing: From Life-Cycle Theory to Housing Tenure Choice

Modern household macroeconomics begins with the Life-Cycle Hypothesis of Modigliani and Brumberg (1954), in which forward-looking individuals plan consumption and saving over their entire lifetime, producing a hump-shaped consumption profile that rises in early working years, peaks in middle age, and declines in retirement as accumulated wealth is drawn down.

Drawing on this theory, the apparatus for embedding different finite-lived agents in macroeconomic equilibrium was developed through the overlapping-generations (OLG) framework. Samuelson (1958) constructed the first OLG model in an exchange economy with three-period lives, and Diamond (1965) extended it to a production economy with capital accumulation. A central limitation of these early models was, however, their deterministic treatment of lifetimes. Yaari (1965) first introduced stochastic lifetimes into consumer theory and showed that without perfect annuity markets agents hold precautionary wealth and leave accidental bequests upon death (formalized in general equilibrium by Abel 1985). Building on this, Blanchard (1985) proposed the perpetual youth model, where agents face a constant, age-independent probability of death. Gertler (1999) captured life-cycle variation by introducing different stochastic transitions between working, retired phases and death, which Grafenhofer et al. (2007) generalized to an arbitrary number of age groups. The multi-stage stochastic-aging structure is at the base of the model used in Kaas et al. (2021), and, through it, of this thesis.

In parallel with the OLG literature, a separate strand studied the consequences of market incompleteness for individual behavior and aggregate outcomes. Bewley (1977) first analyzed a consumer facing stochastic income, borrowing constraints, and no access to insurance markets, showing that the interaction of income uncertainty with borrowing constraints induces precautionary accumulation. Huggett (1993) computed a stationary general equilibrium with a continuum of such agents in an exchange economy, and Aiyagari (1994) extended the framework to a production economy. Deaton (1991) and Carroll (1997) provided complementary theoretical and empirical analysis of buffer-stock saving, showing that liquidity constraints and income uncertainty generate consumption profiles that depart significantly from the frictionless life-cycle prediction.

The synthesis of the OLG and incomplete market traditions was achieved by Huggett (1996), who combined multiple finite-lived generations, uninsurable earnings risk, borrowing constraints, stochastic aging, and accidental bequests in a general equilibrium framework. The model employed in this thesis inherits this integration of incomplete markets with life-cycle heterogeneity.

It is worth observing that the Bewley–Huggett–Aiyagari framework treats household wealth as a single, perfectly liquid asset. In reality, however, housing, the dominant component of household portfolios in most developed economies, departs from this assumption along several dimensions: it is simultaneously a consumption good and an investment asset, so that the quantity consumed and the quantity invested cannot be chosen independently; it is indivisible and subject to a minimum size, which creates a discrete entry barrier; it is illiquid, with substantial transaction

costs that discourage frequent adjustment; and it is typically financed through mortgage debt, introducing leverage and a down-payment constraint that ties homeownership to accumulated savings. These features generate a rich set of interactions with life-cycle behavior and income risk that the literature has explored progressively.

Henderson and Ioannides (1983) established that the housing tenure decision is shaped by two opposing forces. On one side, a rental externality, arising from the inability of landlords to fully charge tenants for the costs of utilization of the dwelling, raises the effective price of renting above that of owning. Against this, housing is a risky asset: if a household's desired consumption of housing services exceeds the quantity it would hold for purely investment purposes, owner-occupying forces it to bear more housing risk than is optimal. Flavin and Yamashita (2002) documented the magnitude of this portfolio distortion, showing that younger homeowners are locked into an inefficiently housing-heavy portfolio that crowds out financial assets. Cocco (2005) formalized the finding in a life-cycle model with income uncertainty, confirming that housing as the dominant asset generates precautionary behavior qualitatively different from models with liquid savings alone. Yao and Zhang (2005) showed that access to a rental market alleviates this tension: households optimally rent when liquidity constraints are severe and transition to ownership as those constraints relax. Housing indivisibility and transaction costs also generate substantially more wealth concentration than a model with liquid assets, because the minimum house size creates a saving-up phase that stratifies households into owners and renters with divergent wealth trajectories (Díaz and Luengo-Prado 2010), a mechanism also present in this thesis.

Another strand of literature has focused on the life-cycle behavior of homeownership. In youth, the down-payment constraint forces households into a savings-accumulation phase before they can purchase (Artle and Varaiya 1978). Ortalo-Magné and Rady (2006) showed that this constraint not only delays first-time purchases but also forces constrained buyers into smaller homes. For older households, proportional transaction costs generate a lock-in effect: Yang (2009) showed that high selling costs prevent elderly homeowners from downsizing when their housing needs decline, explaining the flat/mildly declining homeownership rate observed late in the life cycle. Nakajima and Telyukova (2020) confirmed that the combination of illiquid housing wealth and transaction costs explains why many retirees do not dissave as the frictionless model predicts. Down-payment barriers for the young and transaction-cost barriers for the old thus produce a hump-shaped homeownership profile, reproduced in this thesis through age-dependent down-payment requirements and proportional transaction costs on purchase and sale.

A distinct motivation for homeownership is the desire to hedge against rental-rate uncertainty. Sinai and Souleles (2005) showed that homeownership provides insurance against fluctuations in the cost of housing services: households in areas with more volatile rents are significantly more likely to own, and the value of this hedge increases with the expected duration of tenure. The rent-risk channel is important to this thesis, as private-market renters face idiosyncratic rental-price risk, absent once the household becomes a homeowner. This hedging motive is also the reason for households to access the social housing sector, as tenants enjoy a risk-free,

below-market rent that provides insurance similar to ownership.

2.2 Policy Evaluation

A substantial body of work has used calibrated life-cycle models to study how specific housing policies affect tenure outcomes within a single country. Gervais (2002) showed that the tax-favored treatment of owner-occupied housing in the United States (through untaxed imputed rents and mortgage interest deductibility) distorts both capital accumulation and the homeownership profile over the life cycle. Chambers, Garriga, and Don E Schlagenhauf (2009a) demonstrated that demographic change and mortgage market innovations account for most of the secular rise in U.S. homeownership during the twentieth century. Sommer and Sullivan (2018) find that repealing the U.S. mortgage deduction would raise both homeownership and welfare, because the resulting house price decline more than offsets the loss of the tax subsidy, in contrast to Floetotto, Kirker, and Stroebe (2016). These papers identify mechanisms within a country but do not address the question of why homeownership rates differ across countries.

Cross-country empirical evidence points to a combination of factors. Chiuri and Jappelli (2003) show that the stringency of mortgage down-payment requirements is a significant determinant of homeownership differences across fourteen OECD countries, especially for young households, and Bicakova and Sierminska (2008) confirm this using micro data for five European countries. Beyond financial institutions, transfer taxes, social housing provision (see Sections 2.2.1 and 2.2.2), rental-market regulation, and cultural attitudes (Huber and T. Schmidt 2019) have all been identified as potential explanatory factors.

To explain these differences in a structural model, a growing strand of literature has applied a policy transplant methodology, where a model is calibrated to a benchmark country and selected institutional features are replaced with those of a comparator to isolate their individual and combined effects on homeownership. Wook and S. Cho (2012) calibrate a life-cycle model to South Korea and transplant U.S. housing policies, finding that mortgage market structure and rental arrangements explain about 52 percent of the homeownership differential. Kaas et al. (2021) apply the same methodology to the Germany–U.S. pair, showing that three policy differences (lower RETT, mortgage interest deductibility, and the absence of social housing) close the gap by about two thirds. Kindermann and Kohls (2018) focus on rental-market frictions across European countries and find that differences in the rental-ownership cost wedge are a quantitatively important source of cross-country variation. Grevenbrock, Ludwig, and Siassi (2024) directly compare Germany and Italy, adding co-residence as a third living arrangement, and document that individual-level homeownership rates become very similar for young individuals. Fehr, Hofmann, and Kudrna (2025) focus on pensions and income taxes, finding that pensions have decisive effects on homeownership differences across Germany, Australia, and the United States. Le Blanc, Slacalek, and White (2025) estimate a life-cycle model on micro data for five countries and find that differences in house-price beliefs and the rental wedge are the primary drivers of housing-wealth dispersion.

The present thesis applies the Kaas et al. (2021) transplant methodology to the Germany–Italy

pair, replacing two German policies with their Italian counterparts: a lower real-estate transfer tax, and a social housing system with tighter income targeting, lower rents and a right-to-buy option at a statutory discount. The analysis is also complementary to Grevenbrock, Ludwig, and Siassi (2024), since while they focus on household formation and co-residence, this thesis holds household structure fixed and asks whether institutional settings can account for the homeownership gap. The literature on the two policy instruments at the center of the analysis is reviewed in detail next.

2.2.1 Transfer Taxes

Empirical work has documented significant distortionary effects of real-estate transfer taxes (RETT) on housing-market outcomes. Petkova and Weichenrieder (2017) analyze the German RETT using annual aggregate data and find that a 1 percentage point increase reduces single-family house transactions by approximately 0.23%. Fritzsche and Vandrei (2019) use micro data on German transactions and confirm significant quantity effects. Dolls et al. (2025) find that a 1 percentage point RETT increase reduces property prices by roughly 3% within a year, with larger effects for apartments than for single-family houses. Evidence from outside Germany is consistent: Kopczuk and Munroe (2015) study New York's transfer tax and find large effects on transactions; Best and H. J. Kleven (2018) and C. A. L. Hilber and Lyytikäinen (2017) use notches in the UK stamp duty to document large distortions in both transaction volume and timing.

In quantitative models, Kaas et al. (2021) find that reducing the German RETT from 5.2% to the US level of 0.33% raises the homeownership rate, but at a welfare cost of approximately 0.5% of consumption, because the surge in housing demand pushes up prices. D. J. Schmidt (2022) analyse a revenue-neutral abolition of property transfer taxes in the Netherlands, finding an increase in homeownership but an ambiguous welfare response depending on the replacement instrument. Y. Cho, Li, and Uren (2024) study the welfare implications of the 4% Australian transfer tax and consider reforms that replace the revenue through either a recurring property tax or a consumption tax. Han, Ngai, and Sheedy (2025) develop a search model with endogenous tenure choice calibrated to Toronto and find that higher transaction taxes reduce homeowner mobility and shift demand toward the rental sector.

2.2.2 Social Housing

Social housing interacts with homeownership through two main channels: first, by providing rental housing at below-market rates, it reduces the incentive for eligible households to save for a down-payment and transition into private ownership; at the same time, in countries where social housing tenants are granted a right-to-buy (RTB) option to purchase their unit at a discount, the sector can serve as a direct stepping stone into homeownership.

The relative importance of these two channels varies sharply across countries due to diverse institutional settings. OECD (2020) documents wide variation in social housing stock size (from under 4% in Southern Europe to over 20% in Denmark and Austria), targeting, and eligibility

rules. Consequently, much of the empirical literature relies on country-specific policy reports to understand local tenure decisions.² Overall, empirical evidence on the first channel is well established; for example, C. Hilber (2007) find significant negative effects of public housing provision on private homeownership across European regions. Furthermore, from a broader welfare perspective, Bérard and Trannoy (2025) find that cash housing benefits are generally more cost-effective than these in-kind benefits at reducing poverty and inequality.

Within the quantitative macroeconomic literature, structural analyses of the effect of social housing programs on housing markets are scarce,³ partly because this sector plays a minor role in the US market. Sieg and Yoon (2020) serve as a notable exception: they construct an incomplete-markets framework calibrated to New York City to evaluate the welfare impact of subsidized housing, and demonstrate that expanding public housing yields welfare benefits for the renting population. More relevant to the European context, Kaas et al. (2021) provide the most complete treatment of the German social housing program in a structural model, incorporating income-based eligibility, rationed access, a subsidized risk-free rent, and a size constraint on units. They find that abolishing the program raises both homeownership and welfare across all income deciles.

Despite its prominence in some European countries, the right-to-buy option has also received limited formal economic analysis. Disney and Luo (2017) provide the only structural welfare analysis of the mechanism to date, finding that the RTB option yields ambiguous welfare responses depending on the quality of the unit provided.

To the best of my knowledge, no comparable structural analysis has been conducted for the Italian social housing system or its right-to-buy mechanism. The present thesis addresses this gap by implementing the main features of the Italian social housing system in a life-cycle model and embedding an RTB option that, while relying on some features of Disney and Luo (2017), departs from it in several respects (see Section 5.2). Having established the theoretical and empirical context, the next section describes the institutional differences between Germany and Italy that motivate the counterfactual experiments.

3 Housing Policies in Germany and Italy

The housing environments in Germany and Italy share some similarities but differ along many dimensions, not all of which are equally tractable within the framework chosen for this thesis.

On the one hand, both countries exempt imputed rental income and capital gains on primary residences from income taxation ((OECD 2022; ECB 2003)) and feature conservative mortgage markets with similar loan-to-value ratios (ECB 2009).

On the other hand, Germany's rental market offers structurally stronger long-term security

²See e.g. Baldini, Poggio, et al. (2013) for Italy, Alberdi (2014) for Spain, and Stuart et al. (2015) for the UK.

³A related structural literature studies the allocation side of public housing provision, focusing on waiting lists, targeting, and the trade-off between tenant choice and allocation efficiency: see for instance Geyer and Sieg (2013) for Pittsburgh and Waldinger (2021) for Massachusetts. This work focuses on allocation efficiency within the social housing sector rather than its interaction with homeownership, and is therefore complementary to the present analysis.

for tenants through open ended contracts, strict eviction protections, and since 2015 a rent brake capping new contract rents in tight markets (Bürgerliches Gesetzbuch (2002)). Italy instead reformed its rental market with Legge 431/1998 (Repubblica Italiana 1998b), introducing a free market segment with fixed term contracts alongside a regulated segment where rents are capped by local agreements in exchange for landlord tax incentives. The resulting difference in tenure security and contract flexibility is unlikely to be tractable within the Kaas et al. (2021) framework without substantial extensions, and is thus left for future research.

Furthermore, property taxation differences might distort the buy-versus-rent margin between the two countries. While Germany applies its property tax uniformly across all tenure types, Italy effectively penalizes renting by fully exempting owner-occupied primary residences from the main municipal property tax. However, implementing this asymmetric tax would require either comparing the Kaas et al. (2021)'s benchmark, whose internal parameters are calibrated to an economy without a recurring tax, with a counterfactual where such a tax is introduced, or a recalibration of the German benchmark including recurrent property taxes. Neither option is compatible with the replication logic of the benchmark adopted in this thesis.

There are also differences in the tax treatment of mortgage interest. In Germany, mortgage interest is deductible only on rental properties, not on owner occupied housing. Italy allows a 19 percent deduction on owner occupied mortgage interest capped at 4000 euros per year (Repubblica Italiana 1986). Given the limited magnitude of this cap and the quasi-experimental evidence of Gruber, Jensen, and H. Kleven (2021) that mortgage deductibility affects house size but not the rent versus buy decision, this policy is not implemented as a counterfactual.

The two institutional settings that differ quantitatively and that can be analyzed within this framework, the real-estate transfer tax and the social housing system, are described next.

3.1 Real-Estate Transfer Taxes

A real-estate transfer tax (RETT) is a one-time tax levied on the buyer when ownership of a property changes hands. Because it is paid upfront and in proportion to some form of housing price, the RETT effectively raises the cost of entering homeownership and increases the total transaction costs associated with housing mobility.

The German RETT (Grunderwerbsteuer) was set at a uniform rate of 3.5% of the purchase price across all federal states until 2006. In that year, legislative authority was devolved to the individual states, which subsequently raised their rates. As of 2026, German RETT rates range from 3.5% in Bavaria to 6.5% in Brandenburg, with an average of around 5% (Fritzsche and Vandreì 2019). In addition to the RETT, buyers in Germany face notary fees and real-estate agent commissions (typically shared between buyer and seller). The Italian RETT is the *imposta di registro*, levied upon the transfer of property ownership. Italy's system distinguishes between purchases of a primary residence and other properties. Under Legge 23/12/1999, n. 488 (Repubblica Italiana 1999) the primary residence rate was set at 3% of the property's cadastral value, definable as an administrative value assigned to a property by government land registries for tax purposes, generally different from the market price. A reform enacted through

D. Lgs. 23/2011 (Repubblica Italiana 2011) reduced the rate from 3% to 2% and unified the non-primary-residence rate at 9%. As of 2026, these rates remain in force. Notice that cadastral and market values differ substantially, with cadastral values typically below half of market values (MEF 2012), reflecting the outdated nature of the Italian land registry. The effective tax rate paid by Italian households may therefore even be lower.

3.2 Social Housing

Social housing refers to publicly subsidized rental housing provided to eligible households at rents below the prevailing market level. The German social housing system (Sozialer Wohnungsbau) has its origins in the postwar housing shortage. Eligibility is income based, with the threshold nowadays set slightly below median household income, so that a broad share of the population qualifies (Kirchner 2007). Under this system, rents are legally capped based on costs; Deschermeier, Haas, and Voigtländer (2015) calculate that these regulated rates sit roughly 20 percent lower than comparable private market rents. Eligibility is not reassessed after entry, so households whose income rises above the threshold may continue to occupy social housing at the subsidized rate. Particularly, social housing units are financed through subsidized loans with a certain maturity, after which time they return to the private market (Kaas et al. 2021, p. 133). The social housing stock has declined steadily from 19.4 percent of all residential units in 1968 to 7.1 percent in 2002 (Kirchner 2007) and around 3 percent in 2017 (Kofner 2017), partly reflecting a paradigm shift from supply-side object subsidies towards demand-side housing allowances (such as Wohngeld).

Italy's social housing system (Edilizia Residenziale Pubblica, ERP)⁴ developed along a different trajectory. Public housing units are owned by public entities, and their construction in the 1970s and 1980s was financed primarily through a payroll-based contribution by employers and employees (GesCal). The fund was however frozen in 1998 and no systematic replacement was introduced (Belotti and Arbaci 2021). Access to the program is now governed by regional legislation, following the devolution of housing competence to the Regions under D.Lgs. 31 marzo 1998, n. 112 (Repubblica Italiana 1998a). There is no single national eligibility threshold; instead, each region sets a ceiling for entry based on a wealth indicator called ISEE.⁵ The eligibility thresholds cluster though in a relatively narrow range across regions, ranging from €17,000 to €22,000. While these thresholds can theoretically encompass different households deciles in the first half of the income distribution, the Italian model has become residualistic in its implementation (Poggio and Boreiko 2017), meaning that it has moved towards a position in which it basically provides only a safety net for low-income households (Harloe 1995), also

⁴This represents the traditional publicly-owned rental housing stock in Italy. A broader category, Edilizia Residenziale Sociale (ERS), was introduced by D.M. 22 aprile 2008 to encompass additional forms of subsidized housing provision (cooperatives, foundations, private developers with public support). Since the ERS category postdates most of the calibration window (1995–2014) and the quantitative evidence on stock size, rent discounts, targeting, and the right-to-buy all pertain to the ERP segment, this is the appropriate reference for the counterfactual parameters.

⁵Indicatore della Situazione Economica Equivalente, it represents a composite means-test indicator defined on household income and, partly, on asset ownership.

given lack of sufficient funds to accommodate for higher income households requests (Caruso 2017). The stock composition of the ERP tenants reflects this tendency: Baldini, Poggio, et al. (2013) report that the share of social housing tenants belonging to the poorest income quintile rose from 46% in 1993 to 59% in 2004, with roughly three-quarters of all tenants concentrated in the bottom two quintiles by 2006. However, the remaining quarter of tenants is also distributed across the upper half of the income distribution, a consequence of very low turnover and the absence of effective ongoing means-testing after initial assignment (Poggio and Boreiko 2017).⁶ Italian social housing rents are determined through an income-graduated system established by regional legislation, following Legge 392/1978 (Repubblica Italiana 1978). While no single study provides an estimate of the ERP rent-to-market ratio, several sources point to a non-negligible discount. Poggio and Boreiko (2017) show that between 2008 and 2014 only 17% of ERP tenants in the bottom two income deciles had a rent to income ratio above 20 percent, compared to 70 percent of private renters. Bianchi (2014) reports ERP rents of less than 100 euros per month against market equivalents of roughly 800 euros, implying a discount of about 85 percent. Baldini and Poggio (2012) estimate the total implicit social housing subsidy at approximately 2.7 billion euros per year across 900,000 households. Taken together, the evidence points to an effective discount ranging from as much as 85 percent for the poorest urban tenants to around 30 percent near the income threshold. A final distinctive feature of the Italian system is the right-to-buy mechanism, introduced by Legge 560/1993 (Repubblica Italiana 1993). This legislation gave social housing tenants the option to purchase their unit after a 5-year minimum renting period. It is worth noticing that this option is not a standing right, rather it materializes only when regional governments include a given building in a sale plan, a process over which tenants have no control. If this option is offered, the sale price comes at a huge discount, with estimates of around 68–84% discount relative to market prices (Bianchi 2014). To mitigate immediate speculation, the law imposes a 10-year prohibition on resale following the purchase. Even after this lock-in period, the government retains a permanent right of first refusal, under which it can repurchase the unit at the inflation-adjusted discounted price.⁷ The complexity of the norm, along with Italy's broader cultural orientation toward homeownership promotion, has thus helped making resale unattractive and unpractical even after the lock-in expires. All in all, the RTB option, along with a concurrent reduction in public investment, has been directly responsible for the reduction of the social housing stock, with the OECD (2023) reporting that in 2011 the share of social housing stock represented approximately 4% of the total housing stock, with a further decline in 2020 to only 2.4%.

⁶The available evidence only describes the stock composition at a point in time, not the effective flow of households entering the system.

⁷The owner may alternatively extinguish this right by paying the agency 10% of the current cadastral value, after which free-market sale becomes possible. This, however, comes with an additional administrative and financial cost that varies across regions (See Art. 1, comma 10, 20, 25 of Legge 560/1993)

4 Model

This section presents a baseline open economy model, (following deliberately the same structure and mechanisms as in Kaas et al. (2021)), against which the counterfactual experiments described in Section 5 will take place.

Demographics. Time is discrete and the economy is populated by overlapping generations of households. Each household passes through five age groups indexed by $\tau \in \{1, 2, 3, 4, 5\}$, which correspond to the age brackets 25–34, 35–44, 45–54, 55–64, and 65 and above. The first four groups represent working-age households, while $\tau = 5$ denotes retirement. Transitions between age groups are stochastic⁸. In each period, a household in age group $\tau \leq 4$ advances to group $\tau + 1$ with probability ϑ_τ and remains in its current group with probability $1 - \vartheta_\tau$. A retired household ($\tau = 5$) dies with probability ϑ_5 , in which case it is replaced by a newborn entering at $\tau = 1$. More specifically, every period a $\vartheta_5/(1 + 40\vartheta_5)$ mass of households enters the economy without any assets and each household draws their initial income decile i uniformly from $\{1, \dots, 10\}$ (Kaas et al. 2021, p. 134). This stochastic aging structure generates a stationary age distribution without requiring an explicit calendar-time dimension, while the entry process allows the mass to be constant and normalized to unit.

Income. Labor income $y(\tau, i)$ depends in its log on a age-specific deterministic component M_τ and on its stochastic residual component $\varepsilon_{i,\tau}$, with decile $i \in \{1, \dots, 10\}$. For working-age households ($\tau \leq 4$), the transitions between residual income deciles over time are governed by an age-specific discrete Markov process, denoted by the transition matrix $\Psi_\tau(i'|i)$ (Kaas et al. 2021, p. 134). Persistence in income increases with age, as empirically shown by Storesletten, Telmer, and Yaron (2004). During retirement ($\tau = 5$), households draw a deterministic pension. To capture the link between lifetime earnings and pension wealth, a household's pension decile i maps exactly to their final working-age residual income decile (Kaas et al. 2021, p. 135).

Preferences. Households choose consumption and savings plans so as to maximize expected discounted lifetime utility, with β denoting the subjective discount factor. Period utility is defined over non-housing consumption c and housing services s through a Cobb–Douglas aggregator embedded in a CRRA framework, a specification widely adopted in the quantitative macroeconomic literature on housing:⁹

$$u(c, s; \tau, \mathbb{I}_{h>0}) = \frac{1}{1 - \gamma} \left[(c/n_\tau)^\zeta (\xi_{\mathbb{I}_{h>0}}^\tau s/n_\tau)^{1-\zeta} \right]^{1-\gamma} \quad (1)$$

where $\gamma > 1$ governs the degree of relative risk aversion, and $\zeta \in (0, 1)$ determines the budget share devoted to non-housing goods. To account for changes in household composition across the life cycle, both consumption arguments are scaled by the age-specific equivalence scale n_τ . The term $\xi_{\mathbb{I}_{h>0}}^\tau$ enters as a tenure- and age-dependent shifter on the housing services argument. The

⁸Similar applications in an incomplete market setting with housing by Fernández-Villaverde and Krueger (2011) and Yang (2009).

⁹See e.g. Díaz and Luengo-Prado (2010), Fernández-Villaverde and Krueger (2011), and Sommer and Sullivan (2018).

shifter is normalized to unity for households of working age as well as for retired households who rent. For retired owner-occupiers, however, the parameter is allowed to take values greater than one, capturing the notion that ownership during retirement may carry additional subjective value beyond the pure consumption of shelter (Kaas et al. 2021, p. 135). Bequests arise accidentally from the timing of death. When a household in age group $\tau = 5$ dies, its estates (the sum of financial assets and the after-selling-cost value of housing) are randomly distributed to households in the first two age groups with probability π^I .¹⁰

Housing. Housing assets are $h \in \mathcal{H} \cup \{0\}$, where $\mathcal{H} = [h_{\min}, \infty)$ and $h_{\min} > 0$ is a minimum house size constraint (Cocco 2005; Gervais 2002). This indivisibility means that a household must accumulate sufficient resources to purchase at least $h_{\min}$ units before it can transition into homeownership. Housing units are traded at an endogenous price p , and can be owned either by households or by real-estate firms. Real-estate firms are risk-neutral and operate under perfect competition, acquiring residential units and supplying them to the rental market at the average rate $\bar{\rho}$ (Kaas et al. 2021, p. 135). Owner households allocate their housing stock between personal use and the rental market. Specifically, a household occupies up to its entire stock h as housing services s , while any unused portion $h - s$ can be rented to other households, gaining an average rental rate $\bar{\rho}$; such households are referred to as landlords (Kaas et al. 2021, p. 136).¹¹ When trading housing units, households incur proportional transaction costs: a buyer's cost equal to a fraction t_b of the purchase price and a seller's cost equal to a fraction t_s of the sale price, as in Sommer and Sullivan (2018). Within t_b , a component $\tilde{t}_b \leq t_b$ represents the real-estate transfer tax (RETT). Housing is also subject to idiosyncratic house value risk, following Jeske, Krueger, and Mitman (2013). Given an end-of-period housing unit $\tilde{h}'$, the housing stock next period h' evolve as

$$\log h' = \log(1 - \delta) + \log \tilde{h}' + \chi' \quad (2)$$

where $\delta > 0$ is the annual depreciation rate and $\chi' \sim \mathcal{N}(-\sigma_\chi^2/2, \sigma_\chi^2)$ is an idiosyncratic house value shock.

Financial Assets. Households save in a risk-free asset paying real interest rate r , while they borrow via mortgage loans at rate $r^m > r$. The spread between the mortgage rate and the risk-free rate, $r^m - r$, is taken as given and held fixed throughout, capturing in reduced form the overhead costs that lenders incur per unit of credit extended (Kaas et al. 2021, p. 136). Letting a' denote the household's choice of net financial assets, this is subject to a down-payment constraint:

$$a' \geq -p \tilde{h}' (1 - \theta_\tau) \quad (3)$$

where θ_τ is an age-dependent down-payment requirement (Chiuri and Jappelli 2003).

¹⁰The theoretical foundation originated with Yaari (1965) and was later formalized by Abel (1985). Fernández-Villaverde and Krueger (2011) also adopt a similar structure, but with accidental bequests distributed uniformly across all alive agents, whereas Yang (2009) only to newborn agents.

¹¹This structure follows the framework developed by Chambers, Garriga, and Don E Schlagenhauf (2009a), Chambers, Garriga, and Don E Schlagenhauf (2009b), and Chambers, Garriga, and Don E. Schlagenhauf (2009c) which was subsequently extended by Sommer, Sullivan, and Verbrugge (2013) to include equilibrium prices and rents.

Rental Markets and Social Housing. Households without property wealth ($h = 0$) meet their housing demand s either through the private rental market or, subject to income constraints, through the social housing system (Kaas et al. 2021, p. 137). Households participating in the private rental market face an idiosyncratic unit price ρ for their chosen housing services s . To capture rental market risk, this idiosyncratic rent is modeled as an AR(1) process:

$$\log \rho' = (1 - \omega) \log \bar{\rho} + \omega \log \rho + \nu', \quad \nu' \sim \mathcal{N}\left(-\frac{\sigma_\nu^2}{2(1 + \omega)}, \sigma_\nu^2\right) \quad (4)$$

where the parameter $\omega \in (0, 1)$ dictates the autocorrelation of the rent, and ν' represents a normally distributed shock with volatility σ_ν (Kaas et al. 2021, p. 137).¹² Landlords can diversify idiosyncratic rental risk and therefore receive the average market rent $\bar{\rho}$. Idiosyncratic rent risk provides a hedging motive for homeownership: by purchasing a home, a household eliminates its exposure to future rental rate fluctuations Sinai and Souleles (2005).

Another way to hedge rent risk is through accessing a social housing program. If a household becomes a social housing tenant, it benefits from a below-market rate $\rho^s < \bar{\rho}$, which is deterministic and set by the government (Kaas et al. 2021, p. 137). However, social housing services are subject to an upper size constraint $s \leq \bar{s}$, where $\bar{s}$ represents the largest available social housing unit (Kaas et al. 2021, p. 137). Access to social housing is available only to working-age households ($\tau \leq 4$) and is rationed. A renter household gains access with probability

$$\pi_\tau(y) = \begin{cases} \pi & \text{if } y \leq \bar{y} \text{ and } \tau \leq 4 \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

where $\bar{y}$ is an income eligibility limit and π is a uniform rationing probability. Note that eligibility is assessed only upon entry: a household that has been admitted to social housing can remain even if its income subsequently rises above the threshold. Departure from social housing can arise through three mechanisms: a household may transition into homeownership, which automatically terminates its subsidized tenancy; it may opt to return to the private rental market (for instance, when its desired housing consumption exceeds the size ceiling $\bar{s}$ or when a sufficiently low draw of the idiosyncratic market rent ρ makes private renting comparatively attractive); households face an exogenous displacement shock that terminates social housing access with probability η each period (Kaas et al. 2021, p. 137).

Real-Estate Firms. Real-estate firms incur a per-unit monitoring cost c^m on each unit they rent out.¹³ The zero-profit condition of the real-estate firm links then the equilibrium house price

¹²This means that market rents in the stationary equilibrium are log-normally distributed with mean $\log(\bar{\rho}) - \sigma_\nu^2/[2(1 - \omega^2)]$ and variance $\sigma_\nu^2/(1 - \omega^2)$, so that the unconditional mean of ρ equals $\bar{\rho}$ (Kaas et al. 2021, p. 137).

¹³This parameter captures in reduced form the rental externality formalized by Henderson and Ioannides (1983): because tenants do not internalize the effect of their usage on the property's future value, they tend to over-utilize the dwelling, generating excess depreciation that landlords must monitor and bear. Household landlords are assumed exempt from this cost because they typically reside in close proximity to the rented unit.

and the average rental rate, taking into account this cost (Kaas et al. 2021, p. 138):

$$(r + \delta) p = \bar{\rho} - c^m \quad (6)$$

The real-estate firm acquires regular housing units from a construction sector at the market price p , which is the same price at which such units are traded between households. The firm's zero-profit condition therefore says that the rental income net of monitoring costs, $\bar{\rho} - c^m$, must cover the user cost of capital $(r + \delta)p$ at which the firm holds the unit.

Social housing units are also operated by real-estate firms that commit to renting units at the below-market rate ρ^s with access restricted to eligible households (Kaas et al. 2021, p. 138). A social housing unit reverts to the regular market with probability Φ each period (corresponding to the expiry of the subsidy commitment), at which point it becomes a standard rental unit valued at p . The zero-profit condition for social housing firms is:

$$(r + \Phi + \delta) p^s = \rho^s - c^m + \Phi p \quad (7)$$

with $p^s < p$ being the the social housing unit market price real-estate firms pay to the construction sector. Because social housing units generate a restricted rental yield $\rho^s < \bar{\rho}$ until they revert to the private market, their capitalized market value to the firm, p^s , is strictly lower than the market price p .

Housing Construction. A competitive construction sector produces both regular and social housing units. Building I regular and I^s social housing units in a given period involves a total cost $K(I + I^s)$, where $K(\cdot)$ is an increasing and convex function. The convex structure of the cost function is intended to reflect the limited availability of developable land and the bottlenecks that arise in the construction industry as the pace of building accelerates (Kaas et al. 2021, p. 138).¹⁴ In this model, the construction cost function takes the form:

$$K(I + I^s) = k_0 \frac{(I + I^s)^{1+\phi}}{1 + \phi} \quad (8)$$

where $k_0 > 0$ is a scale parameter and $\phi > 0$ is the inverse of the long-run price elasticity of new housing supply. The construction firm sells regular units to real-estate firms at price p and social housing units at price p^s , while receives an additional per-unit subsidy ς from the government for each social housing unit built. In equilibrium, the construction firm is indifferent between the two output types because the total per-unit revenue from each equals the marginal cost, as evidenced by the profit maximization of the construction firm:

$$p = K'(I + I^s) = p^s + \varsigma \quad (9)$$

¹⁴The underlying mechanism is modeled explicitly by Davis and Heathcote (2005) who show that combining a constant-returns construction technology with land in fixed supply produces an upward-sloping aggregate supply curve for housing. Among others, Floetotto, Kirker, and Stroebel (2016) and Sommer and Sullivan (2018) follow a similar structure for their construction sector modeling choice.

Government. The government uses income taxes (T_{rev}^{inc}) and real-estate transfer tax (T_{rev}^{RE}) to pay for pension payments ($Cost_{pensions}$), construction subsidies for social housing ($Cost_{SH}$) and unspecified public goods G . The tax base y^t aggregates earnings from labor or pensions, interest generated by financial savings, and net rental yields (Kaas et al. 2021, p. 139). Specifically,

$$y^t = y(\tau, i) + r \max[a, 0] + \bar{\rho} \max(0, h - s) - r^m \min\{\max[-a, 0], \max[p(h - s)(1 - \theta_\tau), 0]\} \quad (10)$$

The last term reflects that only landlords may deduct mortgage interest payments, but only up to the lendable fraction $(1 - \theta_\tau)$ of the value of housing that is rented out. The tax function takes the parametric form: $T_\tau(y^t) = y^t - \lambda_\tau (y^t)^{1-\varphi_\tau}$, where φ_τ and λ_τ are age-specific parameters (Kaas et al. 2021, p. 143).¹⁵ When a housing transaction occurs (i.e. the household changes its housing holdings, $\tilde{h}' \neq h$), the buyer pays a transfer tax equal to $\tilde{t}_b \cdot p \cdot \tilde{h}'$ on the new unit.

The Household Problem. The household's state vector is $(\tau, i, \rho, \sigma, a, h)$, where $\sigma \in \{0, 1\}$ indicates social housing access (Kaas et al. 2021, p. 139). Given its observed state, the household's problem unfolds in two stages. First, the household chooses non-housing consumption, housing services, next-period financial and housing assets, and its social housing status $\tilde{\sigma} \in \{0, 1\}$ (conditional on $\sigma = 1$). Second, a sequence of exogenous shocks is realized: housing depreciation, social housing transitions, demographic changes, bequest distributions, and the draws for next period's income and rent. This uncertainty over the next period's state is captured by the expectations operator $\mathbb{E}$. The household solves the recursive problem:

$$V(\tau, i, \rho, \sigma, a, h) = \max_{c, s, a', \tilde{h}', \tilde{\sigma}} u(c, s; \tau, \mathbb{I}_{h>0}) + \beta \mathbb{E} [V(\tau', i', \rho', \sigma', a' + b', h')] \quad (11)$$

subject to the following constraints:

*Budget constraint:*¹⁶

$$c + a' + p \tilde{h}' = y(\tau, i) + [1 + r \mathbb{I}_{a>0} + r^m \mathbb{I}_{a<0}] a + p h + \max(\bar{\rho}(h - s), 0) - \tilde{\rho} s \mathbb{I}_{h=0} - T_\tau(y^t) - \mathbb{I}_{\tilde{h}' \neq h} [t_b p \tilde{h}' + t_s p h] \quad (12)$$

Housing and services constraints:

$$\tilde{h}' \in \mathcal{H} \cup \{0\}, \quad s \geq 0, \quad s \leq h \text{ if } h > 0 \quad (13)$$

Down-payment & House value evolution: as in (3) and (2).

Social housing eligibility: a household can only use social housing ($\tilde{\sigma} = 1$) if it currently has

¹⁵The form follows the parametric specification introduced by Feldstein (1969) and brought into dynamic heterogeneous-agent models by Bénabou (2002). Heathcote, Storesletten, and Violante (2017) popularized then this form in the quantitative macro literature, given its flexibility in nesting both progressive ($\varphi_\tau > 0$) and proportional ($\varphi_\tau = 0$) taxation. λ_τ on the other hand, represents the average tax burden.

¹⁶The indicator $\mathbb{I}_{\tilde{h}' \neq h}$ implies that households cannot adjust their housing size incrementally; any change in housing consumption requires liquidating the entire existing unit (incurring seller cost t_s) and purchasing a new one (incurring buyer cost t_b).

access ($\sigma = 1$) and chooses to consume housing services within the social housing size cap.

$$\tilde{\sigma} \in \{0, 1\} \quad \text{and} \quad \tilde{\sigma} = 0 \quad \text{if} \quad \sigma = 0 \quad \text{or if} \quad s > \bar{s} \quad (14)$$

Rent determination:

$$\tilde{\rho} = \begin{cases} \rho^s & \text{if } \tilde{\sigma} = 1 \\ \rho & \text{otherwise} \end{cases} \quad (15)$$

Rental risk transition: continuing renters face persistent rent risk via the AR(1) process. Owners who sell their house and newborns draw a fresh rent from the stationary distribution.

$$\log \rho' = \begin{cases} (1 - \omega) \log \bar{\rho} + \omega \log \rho + v' & \text{if } h = 0 \text{ (renter)} \\ \sim \mathcal{N}\left(\log \bar{\rho} - \frac{\sigma_v^2}{2(1 - \omega^2)}, \frac{\sigma_v^2}{1 - \omega^2}\right) & \text{if } h > 0 \text{ (owner becoming renter or newborn)} \end{cases} \quad (16)$$

Social housing status transition: renter households without current social housing access ($\tilde{\sigma} = 0$) may gain access with the age- and income-dependent probability $\pi_{\tau'}(y(\tau', i'))$. Social housing tenants ($\tilde{\sigma} = 1$) who remain renters keep their access with probability $1 - \eta$. Households that become homeowners lose social housing status with certainty.

$$\sigma' = \begin{cases} 1 & \begin{cases} \text{with prob. } \pi_{\tau'}(y(\tau', i')) & \text{if } \tilde{\sigma} = 0 \text{ and } h' = 0 \\ \text{with prob. } 1 - \eta & \text{if } \tilde{\sigma} = 1 \text{ and } h' = 0 \end{cases} \\ 0 & \text{otherwise} \end{cases} \quad (17)$$

Bequests: the bequest distribution $B(\cdot)$ is endogenous and coincides in equilibrium with the distribution of estates $a + p(1 - t_s)h$ left by dying households.

$$b' \sim B(\cdot) \quad \text{with prob. } \pi^I \quad \text{if } \tau \in \{1, 2\}, \quad b' = 0 \quad \text{otherwise} \quad (18)$$

Stationary Equilibrium. The model is solved for a stationary equilibrium in the spirit of Huggett (1996), tracking the definition provided by Kaas et al. (2021, p. 141) (see Appendix H.1 for the computational procedure). This equilibrium consists of a household value function V ; associated policy functions for goods consumption $C(\cdot)$, housing consumption $S(\cdot)$, financial and housing assets $A(\cdot)$ and $H(\cdot)$, and social housing status $\Sigma(\cdot)$; a stationary distribution μ over household states $(\tau, i, \rho, \sigma, a, h)$; a bequest distribution $B(\cdot)$; prices $(p, p^s, \bar{\rho})$; investment levels (I, I^s) ; aggregate housing stocks for the private market and social housing $(\bar{H}, \bar{H}^s)$; and a social housing rationing probability π , such that:

1. **Household optimality:** The value function V and policy functions $(C, S, A, H, \tilde{\sigma})$ solve the household's recursive problem (11) subject to constraints (12)–(18).
2. **Real-estate firm optimality:** The zero-profit conditions (6) and (7) hold.
3. **Construction firm optimality:** The first-order condition (9) holds.

4. **Housing market clearing:** All housing units are occupied, i.e. total housing demand equals total supply: $\bar{H} + \bar{H}^s = \int S(\cdot) d\mu(\cdot)$.
5. **Social housing market clearing:** All social housing units are occupied:
 $\bar{H}^s = \int S(\cdot) \mathbb{I}_{\tilde{\sigma}=1} d\mu(\cdot)$.
6. **Stationarity of the distribution:** The distribution μ is invariant with respect to the exogenous shocks (τ, i, ρ, χ) , the social housing transition (17), and the endogenous policy functions for a' and $\tilde{h}'$.
7. **Consistency of the bequest distribution:** $B(\cdot)$ coincides with the distribution of $a + p(1 - t_s)h$ among households with $\tau = 5$.
8. **Stationarity of housing stocks:** The stock-flow conditions hold: $\delta(\bar{H} + \bar{H}^s) = I + I^s$ and $(\Phi + \delta)\bar{H}^s = I^s$.

Notice that the first flow condition says that total construction replaces depreciated housing, and combined with 9, pins down total housing supply $\bar{H} + \bar{H}^s$ at given price p . The second says that the social housing stock is maintained: new social housing construction I^s replaces both depreciated social housing and units that have reverted to the regular market, where total social housing stock $\bar{H}^s$ is determined in equilibrium by reconciling the flow of new entries with the outflows from depreciation, conversion to regular rental at rate Φ and voluntary exits. Given these equilibrium conditions, the stock of owner-occupied housing is $\bar{H}^{ho} = \int \min\{H(\cdot), S(\cdot)\} d\mu(\cdot)$, the stock rented out by landlord households is $\bar{H}^{hr} = \int \max\{0, H(\cdot) - S(\cdot)\} d\mu(\cdot)$, so that total household-owned housing is $\bar{H}^h = \bar{H}^{ho} + \bar{H}^{hr}$. The residual $\bar{H}^e = \bar{H} - \bar{H}^h$ is owned by real-estate firms.

Government budget balance requires that :

$$\begin{aligned}
 G + \underbrace{\int y(5, i) d\mu(5, i, \rho, \sigma, a, h)}_{Cost_{pensions}} + \underbrace{\overbrace{\zeta I^s}^{Cost_{SH}}}_{T_{rev}^{inc}} &= \int T_\tau(y^t(\tau, i, \rho, \sigma, a, h)) d\mu(\tau, i, \rho, \sigma, a, h) \\
 &+ \underbrace{\tilde{t}^b p \int H(\tau, i, \rho, \sigma, a, h) \mathbb{I}_{H(\tau, i, \rho, \sigma, a, h) \neq h} d\mu(\tau, i, \rho, \sigma, a, h)}_{T_{rev}^{RE}} \quad (19)
 \end{aligned}$$

Calibration. The baseline calibration targets the German economy and is taken entirely from Kaas et al. (2021, pp. 144, 145). All income and wealth values are expressed in thousand euros at 2006 prices. The model period is one year. Parameters are divided into two groups: those set externally from data or the existing literature, and those calibrated internally by requiring the model to match selected data targets. Tables 2 and 3 report the full set of parameter values. Demographic characteristics are either obtained from the German Socio-Economic Panel data (SOEP) or from the structure of the model. The labor income process and the income tax functions are estimated from the SOEP 1995-2015 data following the nonparametric procedure

Table 2: Externally calibrated parameters (Kaas et al., 2021, Table 1).

Parameter	Value	Source / Explanation
<i>Demographics</i>		
– Household size $(n_1, \dots, n_5)$	(1.41, 1.74, 1.70, 1.44, 1.39)	OECD modified equivalence scale (SOEP)
– Ageing prob. $\vartheta_1, \dots, \vartheta_4$	0.1	10-year age groups
– Death prob. ϑ_5	0.05	20-year expected retirement
– Inheritance prob. π^I	0.05	By construction of the mass of households
<i>Preferences and financial markets</i>		
– Risk aversion γ	2	Standard value
– Expenditure share ζ	0.717	Statistisches Bundesamt (2016)
– Real interest rate r	0.0255	Average 1991–2014, Bundesbank
– Mortgage rate r^m	0.0374	Average 1991–2014, Bundesbank
<i>Housing</i>		
– Down-payment $\theta_1, \theta_2, \theta_3$	0.20	Chiuri and Jappelli (2003)
– Down-payment (θ_4, θ_5)	(0.60, 1.00)	No new mortgage in retirement
– Transaction costs $(t_b, \tilde{t}_b, t_s)$	(0.108, 0.052, 0.029)	Pop.-weighted German state averages
– Depreciation δ	0.01	100-year housing lifespan
– Minimum size $h_{\min}$	80	$\approx$ 10th pctile of housing wealth (SOEP)
– Supply elasticity ϕ	2.34	Caldera and Johansson (2013, Table 2)
– House value risk σ_χ	0.104	SOEP, Kaas et al. (2021, Appendix A)
<i>Rental market and social housing</i>		
– Rent persistence ω	0.404	SOEP, Kaas et al. (2021, Appendix A)
– Rent volatility σ_v	0.094	SOEP, Kaas et al. (2021, Appendix A)
– SH rent discount $\rho^s / \bar{\rho}$	0.80	Deschermeier, Haas, and Voigtländer (2015)
– SH eligibility $\bar{y}$	37.8	Median HH labor income; Kirchner (2007)
– Transformation rate Φ	0.04	Schier and Voigtländer (2016)
<i>Income and taxes</i>		
– Income process $(M_\tau, E_\tau, \Psi_\tau)$	—	SOEP, Kaas et al. (2021, Appendix A)
– Tax parameters $(\lambda_\tau, \tau_\tau^e)$	—	SOEP, Kaas et al. (2021, Appendix A)
– Gross replacement rate	0.42	OECD (2011)

described in Kaas et al. (2021, Appendix A). Pension income is set at a 42% gross replacement rate (OECD 2011), with the top and bottom deciles capped at the statutory maximum and minimum annual pensions. Down-payment requirements for working-age households follow Chiuri and Jappelli (2003). Total buyer transaction costs of 10.8% include the RETT of 5.2% (a population-weighted average of German state-level rates, Fritzsche and Vandrei (2019)) plus solicitor and brokerage fees. The long-run price elasticity of new housing supply in Germany is estimated at 0.428 by Caldera and Johansson (2013), implying $\phi = 2.34$.

The seven internally calibrated parameters are identified jointly by solving the model to match data targets from the SOEP (wave 2002, 2007 and 2012). The discount factor β is pinned down by average household wealth; the monitoring cost c^m identifies the aggregate homeownership rate;¹⁷ the retirement preference shifter $\xi_{\mathbb{I}_{h>0}}^5$ matches the homeownership rate among retirees; the social housing parameters $(\bar{s}, \zeta, \eta)$ are identified by the size distribution of social housing units, the social housing stock share of 7.1% (Kirchner (2007)), and the empirical move-in rate; and the cost scale k_0 enforces the normalisation $p = 1$. Kaas et al. (2021) document the fit of this model in their Figures 1–2 and Appendix B. The model reproduces fairly well the age profiles

¹⁷Notice that the homeownership target (42.2%) differs from what is retrieved by Kaas et al. (2021) in the wealth modules of SOEP (circa 44%). This “is the result of the age-specific homeownership rates aggregated according to the population shares of each age group in the model” (Kaas et al. 2021, Appendix A.7, page 7).

of homeownership, net wealth, and its components, as well as the cross-sectional variation of homeownership by income and wealth decile. Mentioned limitations include: an underestimation of homeownership among young lower-middle wealth households, explainable by no direct housing bequests or inter-vivo transfers, and an overprediction of net-wealth accumulation during working life and decumulation during retirement, likely caused by differences in housing size owned by retirees between data and model. These discrepancies are also inherited by the counterfactual experiments (see Section 6.5).

Table 3: Internally calibrated parameters (Kaas et al., 2021, Table 2).

Parameter	Value	Target	Model / Data
Discount factor β	0.9485	Average wealth (€ '000)	128.7 / 128.7
Monitoring cost c^m	0.0189	Homeownership rate (%)	42.5 / 42.2
Owner 65+ utility $\xi_{Ih>0}^5$	1.37	HOR 65+ (%)	47.4 / 47.6
SH max. size $\bar{s}$	212	99th pctl ratio $s/\bar{s}$	0.726 / 0.731
SH subsidy ς	0.1442	SH stock share	0.071 / 0.071
SH exit rate η	0.0155	SH move-in rate	0.0128 / 0.0128
Cost scale k_0	0.2898	Price normalisation $p = 1$	—

5 Counterfactual Experiments

In this thesis I aim to quantitatively test whether differences in homeownership between Germany and Italy can be explained by different institutional factors. I thus conduct the following four counterfactual experiments (CI1–CI4), which replicate some of the most defining features of the Italian housing policy compared to the German one. These experiments are structured as a decomposition exercise that isolates the effect of specific housing policy parameters on homeownership, holding the broader environment constant.¹⁸ A direct implication of this principle is that the social housing stock is not targeted but instead adjusts endogenously to each policy environment, so that the measured homeownership response captures both the direct effect of the policy on household tenure decisions and any indirect effect operating through the resulting change in social housing provision. This means, effectively, that the per-unit social housing construction subsidy ς and the access probability π are held at their benchmark values across all counterfactuals. The baseline economy is calibrated to a window centred on the mid-2000s, early 2010s. The parameter choice for the counterfactual experiments thus adapts to the same time frame, preserving consistency with the German benchmark and following standard practice in the cross-country structural literature.

5.1 CI1

The real-estate transfer tax (RETT) is set to a market-adjusted version of the pre-D.Lgs. 23/2011 reform, in line with the calibration period defined above. MEF (2012) show that in 2010 (and

¹⁸Observe for instance that, as evidenced by Chiuri and Jappelli (2003), the Italian housing supply is particularly inelastic ($\varphi = 0.258$). If this were used, the direct effect of institutional policies could not be isolated, as a lower elasticity would amplify price responses to demand shocks.

similarly in 2012 and 2014) the average ratio between the estimated market price and the potential taxable value (i.e., the cadastral value) of primary housing units across Italy is 2.25. This means that, while the RETT in Italy is 3% on the cadastral value, the effective market-price tax is $\tilde{t}^b = 1.33\%$. Since the RETT enters the budget constraint of the household as an expense tied to purchasing a new home, a reduction in its value reduces the effective cost of transitioning from renting to owning, making the purchase option more attractive for marginal households (Kaas et al. 2021; Y. Cho, Li, and Uren 2024; Han, Ngai, and Sheedy 2025). As described above, the social housing stock adjusts endogenously, so the response to the RETT reduction includes both the direct rent-versus-buy effect and any voluntary exit from the social housing sector induced by cheaper private-market purchase.

5.2 CI2

This counterfactual implements the main defining features of the Italian right-to-buy (RTB) mechanism within the baseline social housing system, so as to isolate its independent effect on homeownership. In the baseline model, the transition from renting to owning takes place solely in the private market, where a household is required to satisfy the down-payment constraint through accumulated financial assets, which creates a saving-up phase (Artle and Varaiya 1978). Including a right-to-buy mechanism offers an additional pathway to homeownership, as now social housing tenants are offered a price that is a fraction of the market value, enabling households that could not afford a private-market purchase to become owners.

Disney and Luo (2017) provide the closest modeling basis for the RTB. However, the present thesis departs from it along three dimensions. In Disney and Luo (2017)’s framework, tenants benefit from a discounted price on their housing unit, the offer to do so is a standing right and the resale restriction takes the form of a repayable discount. Here, social housing tenants can buy any house up to the maximum social housing unit size; the RTB opportunity is not a standing right but an exogenous shock controlled by regional authorities, and the resale restriction is modeled as permanent rather than as a repayable discount. These choices reflect both differences between the Italian and the British systems¹⁹ and computational trade-offs explained below.

Variable Definition. As described in Section 3, households are legally eligible to purchase their social housing unit after a 5-year minimum tenancy, but the offer is exogenous to the tenant because regional authorities decide which units enter the sale plan. I model this by introducing an i.i.d. opportunity shock $\chi^{RTB} \in \{0, 1\}$, where $\chi^{RTB} = 1$ means the household is allowed to purchase and $\chi^{RTB} = 0$ means it is not. Assuming $\Pr(\chi^{RTB} = 1) = \pi_{RTB} = 0.2$ matches a 5-year expected waiting period for initial eligibility. Since χ^{RTB} is i.i.d., its current realisation carries no information about future realisations and therefore does not enter the household’s state vector as a persistent state variable (Stokey, Lucas, and Prescott 1989).

Social housing tenants are assumed to know that the RTB mechanism exists, so the option value of a future offer affects their consumption-saving decisions before any option materialises.

¹⁹See Disney and Luo (2017), Forrest, Gordon, and Murie (1996), and Jones and Murie (2008) for a description of the RTB mechanism in the UK.

This generates two competing effects on tenant behavior. On one side, a wealth accumulation motive: knowing that a discounted purchase opportunity may arrive, tenants save more to be in a financial position to exercise when the offer comes (Dixit and Pindyck 1994). On the other side, an insurance effect (Sinai and Souleles 2005): the RTB option raises the value of social housing tenancy, making it more attractive relative to private-market alternatives and potentially slowing voluntary exit.

If the household decides to purchase, it benefits from a statutory price discount $d = 0.68$ (at the lower bound of the evidence by Bianchi (2014)²⁰), and can choose to purchase a unit up to the maximum social housing size $\tilde{h}' \leq \bar{s}$ (see Appendix C for clarification). The household's balance sheet then records a housing asset worth $p\tilde{h}'$ against an expenditure of only $p\tilde{h}'(1 - d)$. The difference, $d \cdot p\tilde{h}'$, constitutes a discrete net worth gain that can shift the household across several deciles of the wealth distribution.

As seen in Section 3, Italian law imposes a 10 years ban on resale, after which other restrictions apply. Modeling a finite-period restriction mechanism would imply computational costs that might go beyond the scope of this thesis and simultaneously not capture the full set of restrictions that are imposed after the resale ban is lifted (see Appendix D). I thus adopt a conservative specification in which the resale restriction is permanent: once a household exercises the RTB option, it retains the unit indefinitely and cannot sell, resize, or sub-let it (they are thus in a "locked-in" state). This overstates the illiquidity of RTB-purchased properties, depressing the value of being a RTB owner and the take-up rate of the option.

The permanent resale restriction also has a direct consequence for the household's ability to borrow. A mortgage is a secured loan, and if the borrower is legally barred from transferring the property and the asset generates no pledgeable rental income (since $s = h$), the housing unit has zero collateral value. Locked-in homeowners therefore face a non-negativity constraint on financial assets, $a' \geq 0$. The same logic applies at the moment of purchase: no rational lender would extend a mortgage against an immediately non-transferable property, so RTB purchasers face a self-financing constraint requiring them to fund the full discounted price from accumulated liquid assets.²¹

In order to signal whether a social housing tenant exercises the RTB option, I define a RTB indicator $\mathbb{I}^{RTB}$, such that it equals 1 if a social housing tenant receives an offer and its end of

²⁰The estimates provided in the literature are not clear on which unit type they refer to, its quality and its size. It thus might as well be that the upper bound of discounts provided might apply for particularly old buildings located in areas where the gap between cadastral value and market price is large, and not necessarily for the whole units population. Setting $d = 0.68$ represents a lower-bound estimate of RTB's impact on homeownership. See Appendix F for a sensitivity analysis with $d = 0.75$ and $d = 0.84$.

²¹While Legge 560/1993 does provide for installment purchases with mortgage inscription (comma 12(b)), this mechanism is a vendor financing arrangement between the buyer and the selling public entity, whose economic rationale rests on the finite nature of the resale restriction and the agency's ability to recover the property upon default. Under the permanent lock-in adopted here, this recovery mechanism has no counterpart, making the self-financing constraint the internally consistent choice.

period housing size is non-zero:

$$\mathbb{I}^{RTB} = \begin{cases} 1 & \text{if } \tilde{\sigma} = 1, \chi^{RTB} = 1, \text{ and } \tilde{h}' \in (0, \bar{s}] \\ 0 & \text{otherwise} \end{cases} \quad (20)$$

I then define a single binary state variable $L \in \{0, 1\}$, such that $L = 1$ represents the household owning an RTB-purchased unit subject to the permanent resale restriction, while $L = 0$ represents all other types of households (private homeowners, private renters and social-housing renters). Since the lock-in is permanent, the law of motion is:

$$L' = \begin{cases} 1 & \text{if } \mathbb{I}^{RTB} = 1 \quad (\text{household exercises the RTB option}) \\ 1 & \text{if } L = 1 \quad (\text{lock-in persists permanently}) \\ 0 & \text{otherwise} \quad (\text{private owners, private renters, and social renters}) \end{cases} \quad (21)$$

Once $L = 1$, the lock-in persists, regardless of household's decisions, until the household exits the economy through death or its housing unit is destroyed by a depreciation shock. $L = 1$ therefore constitutes an absorbing state: a subset of the state space from which there is zero probability of transitioning back to the complementary states (see e.g. Ch. 2, Ljungqvist and Sargent 2018). This means that, because a locked-in household will never again occupy a transient state ($L = 0$), its continuation value depends only on the value function within the absorbing class itself. The Bellman equation for locked-in households is therefore self-contained and is independent of the value functions for the other states. For consistency with the rest of the framework, once the household dies, RTB units are liquidated at market price upon the holder's death and pooled with financial assets in the bequest distribution, in the same way as any other housing asset.²²

Recursive Formulation. The household population is partitioned into three groups, each with a distinct value function. For notation purposes, I indicate the next period value function V with state x : $V(x') \equiv V'$.

Standard Households ($\tilde{\sigma} = 0, L = 0$): These include all market renters, market homeowners, and former RTB owners whose housing unit has been destroyed. Their value function is exactly as in the baseline model.

$$V^0(\tau, i, \rho, \sigma, a, h) = \max_{c, s, a', \tilde{h}', \tilde{\sigma}} \left\{ u(c, s; \tau, \mathbb{I}_{h>0}) + \beta \mathbb{E}[V^0(\tau', i', \rho', \sigma', a' + b', h')] \right\} \quad (22)$$

Social Housing Tenants ($\tilde{\sigma} = 1, h = 0$): The value function for social housing tenants incorporates

²²In reality, RTB-purchased units can be transferred to heirs, with some constraints, as explicated for instance by art. 12 of L.R. 12/1999 (Regione Lazio 1999). These features would in principle affect how a tenant evaluates the lock-in state, since part of the discount could be recovered by the next generation in the form of inherited housing wealth subject to residual restrictions. The present model abstracts from this channel, however, because bequests are accidental and there is no warm-glow motive in the utility specification: the household derives no value from the size or composition of the estate it leaves behind, so the post-mortem treatment of the unit does not feed back into the lock-in valuation.

the possibility of a future RTB offer:

$$V^1(\tau, i, \rho, \sigma, a, h) = \max_{c, s, a', \tilde{h}'} \left\{ u(c, s; \tau, \mathbb{I}_{h>0}) + \beta \mathbb{E}[V^1(\tau', i', \rho', \sigma', a' + b', h')] \right\} \quad (23)$$

where next period's value function $V^{1'}$ replaces the baseline social housing exit structure to include the RTB option value. In the baseline model, $V^{1'}$ takes the form:

$$V^{1'}_{\text{baseline}} = (1 - \eta) \cdot \Omega' + \eta \cdot V^{0'} \quad (24)$$

where η is the exogenous social housing exit rate and $\Omega \equiv \max(V^1, V^0)$ the endogenous exit choice between staying in the social housing program (V^1) and exiting to the private market (V^0). If the RTB option materialises, the tenant now chooses among three alternatives, written as: $\Omega^{RTB} \equiv \max(V^{RTB}, \Omega) = \Omega + \max(V^{RTB} - \Omega, 0)$. The term $\max(V^{RTB} - \Omega, 0)$ is the RTB option value (Dixit and Pindyck 1994), i.e. the non-negative gain from receiving an offer, conditional on exercising.

Here, the RTB offer arrives with probability π_{RTB} conditional on retaining social housing access. $V^{1'}$ then becomes:

$$V^{1'} = (1 - \eta) \left[\underbrace{(1 - \pi_{RTB}) \cdot \Omega'}_{\text{no offer}} + \underbrace{\pi_{RTB} \cdot \Omega^{RTB'}}_{\text{offer arrives}} \right] + \eta \cdot V^{0'} \quad (25)$$

The exogenous exit term $\eta \cdot V^{0'}$ is unchanged: a household that loses social housing access cannot receive an RTB offer. Using the additive decomposition $\Omega^{RTB} = \Omega + \max(V^{RTB} - \Omega, 0)$, this simplifies to:

$$V^{1'} = \underbrace{(1 - \eta) \cdot \Omega' + \eta \cdot V^{0'}}_{\text{baseline}} + \underbrace{(1 - \eta) \cdot \pi_{RTB} \cdot \max(V^{RTB'} - \Omega', 0)}_{\text{RTB option value}} \quad (26)$$

RTB Value: The value of exercising RTB is:²³

$$V^{RTB}(\tau, i, a) = \max_{c, s, a', \tilde{h}'} \left\{ u(c, s; \tau, \mathbb{I}_{h>0}) + \beta \mathbb{E}[V^{lock}(\tau', i', a' + b', h')] \right\} \quad (27)$$

subject to: $\tilde{h}' \in \{h \in \mathcal{H} : 0 < h \leq \bar{s}\}$, $s \leq \bar{s}$, the RTB budget constraint (32), and the RTB borrowing constraint (35). The continuation value is V^{lock} (the permanently locked-in value function defined below), reflecting the statutory resale restriction activated upon purchase.

RTB Homeowners Under Permanent Lock-in ($\tilde{\sigma} = 0$, $h > 0$, $L = 1$): Since the resale restriction is permanent, the locked-in value function is self-contained:

$$V^{lock}(\tau, i, a, h) = \max_{c, a'} \left\{ u(c, s; \tau, \mathbb{I}_{h>0}) + \beta \mathbb{E}[V^{lock}(\tau', i', a' + b', h')] \right\} \quad (28)$$

²³Since the tenant exits social housing upon purchase, neither the idiosyncratic rent ρ nor the social housing status σ affects the RTB value.

subject to: $\tilde{h}' = h$ (households keep the same unit); $s = h$ (households cannot sub-let during the lock-in period); the locked-in budget constraint (33); and the locked-in borrowing constraint (36).

When the RTB opportunity arrives ($\chi^{RTB} = 1$), the tenant evaluates whether to exercise it by comparing the value of being a locked-in household with the value of remaining a social housing tenant. Therefore, the optimal take-up decision rule that characterizes the choice $\tilde{h}' \in (0, \bar{s}]$ is given by the policy indicator:

$$\mathbb{I}^{RTB}(\tau, i, \rho, \sigma, a, h) = \mathbb{I}\{V^{RTB}(\tau, i, a) > V^1(\tau, i, \rho, \sigma, a, h)\} \quad (29)$$

Budget Constraints. The three groups face also different budget constraints.

Standard Budget Constraint: For $\tilde{\sigma} = 0$, $L = 0$ households, the budget constraint is exactly as in the baseline model:

$$\begin{aligned} c + a' + p \tilde{h}' = & y(\tau, i) + [1 + r \mathbb{I}_{a>0} + r^m \mathbb{I}_{a<0}] a + p h \\ & + \max(\bar{\rho}(h - s), 0) - \rho s \mathbb{I}_{h=0} - T_\tau(y^t) - \mathbb{I}_{\tilde{h}' \neq h} [t_b p \tilde{h}' + t_s p h] \end{aligned} \quad (30)$$

Social Housing Tenant — No RTB: For $\tilde{\sigma} = 1$, $h = 0$ households not exercising RTB:

$$c + a' + p \tilde{h}' (1 + t_b \mathbb{I}_{\tilde{h}'>0}) = y(\tau, i) + (1 + r \mathbb{I}_{a>0} + r^m \mathbb{I}_{a<0}) a - \rho^s s - T_\tau(y^t) \quad (31)$$

Social Housing Tenant — Exercising RTB: For $\tilde{\sigma} = 1$, $h = 0$ households exercising RTB:

$$c + a' + p \tilde{h}' (1 - d)(1 + t_b) = y(\tau, i) + (1 + r \mathbb{I}_{a>0} + r^m \mathbb{I}_{a<0}) a - \rho^s s - T_\tau(y^t) \quad (32)$$

The tenant pays the discounted price plus transaction costs computed on the discounted price. There is no selling cost since $h = 0$. During the current period, the tenant still pays the social rent $\rho^s s$ for housing services consumed.

Locked-in Homeowner: Since $\tilde{h}' = h$ and $s = h$, all housing-related flows cancel:

$$c + a' = y(\tau, i) + (1 + r \mathbb{I}_{a>0} + r^m \mathbb{I}_{a<0}) a - T_\tau(y^t). \quad (33)$$

The locked-in owner's resources consist exclusively of labor or pension income and returns on financial assets. Taxable income simplifies correspondingly to $y^t = y(\tau, i) + r \max[a, 0]$, with no rental income and no landlord mortgage deduction.

Borrowing Constraints. The permanent resale restriction also has a direct consequence for the household's ability to borrow.

Standard Households: As in the baseline model:

$$a' \geq -(1 - \theta_\tau) \cdot p \tilde{h}' \quad (34)$$

RTB purchase: At the moment of the RTB purchase, given the house's unsalability, the household

faces a self-financing constraint:

$$a' \geq 0 \quad (35)$$

Since the household must fund the full discounted price from its liquid wealth and cannot leverage mortgage debt, the maximum affordable house size is bounded by the household's accumulated assets.

Locked-in homeowner: During the lock-in state, for the same reason as above:

$$a' \geq 0 \quad (36)$$

The RTB option thus presents the tenant with a tradeoff. Exercising the option converts a flow-financed housing arrangement into a stock-financed one, eliminating future rental payments but absorbing a large share of the household's liquid assets that cannot be rebuilt through mortgage borrowing. The affordable house size under RTB might also be smaller than the housing services the tenant consumed at the subsidized rent, because the buyer finances the full discounted unit without mortgage support. Third, the permanent lock-in eliminates the possibility of adjusting housing size, a restriction that is particularly costly for young households who face decades of potential life-cycle changes. Together, these features produce a conservative specification of the Italian RTB mechanism.

Government Budget Constraint. The right-to-buy discount generates a capital loss for the real-estate firm operating the social housing unit: the household pays only $p\tilde{h}'(1-d)$ for a unit whose market value (if not sold and reverted to the private market) is $\tilde{h}'p$. To preserve the firm's zero-profit condition (7), the government compensates the firm by paying $d \cdot p\tilde{h}'$ per RTB transaction. The firm's total receipt upon an RTB purchase is therefore $p\tilde{h}'$, the same as upon purchase after reconversion to the regular market at rate Φ . Notice that the RTB transaction reclassifies a unit from the social housing sector to the owner-occupied private market without altering total housing services, so the aggregate housing market-clearing condition is unchanged. The stock-flow equation for social housing also holds in its original form (and with it also the firm zero-profit condition, see Appendix B) but the equilibrium level of $\bar{H}^s$ (assuming no stock replacement) falls considerably, since RTB introduces an additional permanent exit channel. The only aggregate consequence is an additional fiscal cost $Cost_{RTB} = \int d \cdot p \cdot \tilde{h}'_{RTB} d\mu$, where $\tilde{h}'_{RTB}$ is the housing unit purchased under the RTB option. This cost enters the unified government budget:

$$G + Cost_{Pensions} + Cost_{SH} + Cost_{RTB} = T_{rev}^{inc} + T_{rev}^{RE} \quad (37)$$

Social Housing Stock. As noted at the beginning of this section, the access probability π and the construction subsidy ς are held at their benchmark values, so the social housing stock adjusts endogenously to the RTB-induced outflow of tenants. This choice is motivated by the fact that modifying π would itself constitute a policy intervention beyond the introduction of the RTB mechanism. Holding the stock at its benchmark value of 7.1% would in fact require raising π from 1.28% to approximately 10%, a nearly eightfold increase (see Appendix E). Such a regime would thus conflate the effect of the RTB mechanism with that of a much more generous social

housing intake.²⁴ This approach is also consistent with Disney and Luo (2017), where they also let the council housing stock deplete endogenously as tenants exercise their option.

The endogenous specification also simplifies the fiscal implications. Since the construction subsidy ς is held at its benchmark value, the government does not bear the additional cost of replenishing the stock. The only new fiscal burden is the RTB discount, Cost_{RTB} , partially offset by the reduced steady-state expenditure on social housing construction as the stock contracts.

5.3 CI3

This experiment implements the three distinctive design features of the Italian social housing system discussed in Section 3: a deeper rent discount, restricted eligibility, and a right-to-buy mechanism. The aim is to assess whether their joint operation produces homeownership effects that differ from the RTB mechanism in isolation.

Restricted Targeting. In the absence of direct evidence on the income distribution of new allocations, I model the residualistic orientation of the Italian system by restricting eligibility to the bottom three deciles of the income distribution.²⁵ Formally,

$$\pi_{\tau} = \begin{cases} \pi & \text{if } y \leq y_3 \text{ and } \tau \leq 4 \\ 0 & \text{otherwise} \end{cases} \quad (38)$$

where y_3 is the income threshold corresponding to the third decile of the benchmark income distribution. Two effects are expected. Middle-income households who would have qualified under the German rules are now forced to rent on the private market, where the standard rent-vs-buy tradeoff applies and strengthens their saving motive for homeownership. At the same time, a smaller (and poorer) eligible pool reduces the volume of tenure conversions through the RTB channel.

Social Housing Stock. The access probability and the construction subsidy remain at their benchmark values, so the social housing stock adjusts endogenously. Relative to CI2, two forces compound: the RTB-induced contraction documented above, and the tighter eligibility threshold, which restricts the inflow of new tenants, while keeping the per-household access probability the same.

Rent Discount. In the absence of a single authoritative estimate, I set $\rho^s = 0.6\bar{\rho}$, which corresponds to a 40 percent discount relative to the market rent and is consistent with the conservative average implicit subsidy estimated by Baldini and Poggio (2012).²⁶ A deeper

²⁴In reality, Italy's decline of social housing stock occurred over a transition spanning several decades, during which some limited replacement construction did take place. The Italian trajectory would therefore sit between the two specifications considered here.

²⁵Results of an alternative counterfactual where the threshold is set at the second income decile are reported in Appendix F.

²⁶The uniform discount follows the structure of Kaas et al. (2021) and abstracts from the income-graduated schedule used in practice, where poorer tenants receive deeper subsidies; 0.6 represents an average across the eligible population. Appendix F reports robustness results for $\rho^s = 0.2\bar{\rho}$ and $\rho^s = 0.4\bar{\rho}$, which span the plausible range discussed in Section 3.

Table 4: Italian counterfactual parameters.

Parameter	Value	Source
<i>CI1: Real-estate transfer tax</i>		
– RETT $\tilde{\tau}_b$	0.0133	Legge 23/12/1999, n. 488; MEF (2012)
<i>CI2: Right-to-Buy (Riscatto)</i>		
– Purchase discount d	0.68	Bianchi (2014)
– RTB shock π_{RTB}	0.20	Legge 560/1993
<i>CI3: Italian social housing system</i>		
– SH rent discount $\rho^s / \bar{\rho}$	0.60	See text
– SH targeting $\pi_\tau(y)$	First 3 income deciles	See text
– Purchase discount d	0.68	Bianchi (2014)
– RTB shock π_{RTB}	0.20	Legge 560/1993
<i>CI4: Full Italian system</i>	All parameters from CI1 and CI3.	

discount raises the value of remaining a social housing tenant relative to exercising the RTB, especially for households whose liquid wealth is insufficient to exercise the purchase opportunity when it arrives.

The combined design produces a social housing sector that operates differently along the wealth dimension. The poorest eligible households, for whom the rent discount is most valuable and the borrowing constraint is most likely to bind, remain as tenants. The relatively wealthier households within the eligible pool exercise the RTB option and exit into homeownership. Social housing thus should converge toward a residualistic safety net for households genuinely unable to access ownership.

5.4 CI4

The reduced real-estate transfer tax (CI1) and three main features of the Italian social housing system (CI3) are implemented simultaneously. This experiment captures the joint effect of the two policy channels that differentiate the Italian housing system from the German one within the scope of this model. Consistently with the other sections, I let the social housing stock endogenously adjust. The computational strategy of the model under these experiments is described in Appendix H.2.

6 Results

In this section I present the quantitative findings of the four Italian counterfactual experiments relative to the benchmark model, through steady-state comparisons. To disentangle the channels through which each policy affects homeownership, results are reported under four equilibrium concepts. Partial equilibrium (PE) holds both house prices and income taxes at benchmark values, isolating the pure demand-side response. The price-adjust specification lets the house price clear the housing market with taxes fixed. The tax-adjust specification scales by the same proportion for all age groups the income tax parameter λ_τ to restore government budget balance at fixed prices (see Heathcote, Storesletten, and Violante (2017)). General equilibrium (GE) allows both to adjust simultaneously, clearing the housing market and the government budget jointly.

6.1 CH

Reducing the effective RETT from 5.2% to 1.33% raises the homeownership rate by 7.2 pp to 49.7% in general equilibrium (Table 5), closing 27.6% of the Germany–Italy gap. This makes the RETT channel the single most important individual policy lever, with relative magnitudes consistent with the findings of Kaas et al. (2021) for the US and of D. J. Schmidt (2022) for the Netherlands.²⁷

In PE, homeownership rises to 51.2%, an increase of 8.7 pp (Table 6). The gains concentrate in the middle of the wealth distribution (Figure 2, top left). While the poorest deciles lack the accumulated savings to overcome the minimum house size and down-payment constraints regardless of transaction costs, middle-wealth households sit on the margin where the reduced tax wedge makes the jump into ownership affordable. Along the life cycle, the 45–64 age brackets benefit the most (Figure 1, top left): these are the households with sufficient assets to meet the down-payment requirement but for whom the transaction cost wedge previously tipped the tenure decision toward renting. Young households respond more moderately, reflecting binding borrowing constraints, and retirees gain only +5.7 pp, constrained by the rigid down-payment parameters. The portfolio shifts toward housing, with financial assets falling and mortgage debt expanding, as lower transaction costs make housing a more attractive store of value. Aggregate housing demand rises most sharply among the lowest income quintiles, for whom the reduced tax friction provides a proportionally larger easing of their tight budget constraints. Concurrently, the social housing stock contracts endogenously as tenants exit the sector to purchase larger private units at the reduced transaction cost.

When the house price is allowed to adjust, it rises by 1.9% and the price-to-rent ratio increases to 18.50 (Table A1).²⁸ Renting becomes relatively cheaper than buying, partially undoing the direct benefit of the lower RETT. Homeownership reaches 49.2% rather than 51.2%, with demand gains for lower-income quintiles roughly halved since these households are the most price sensitive.

The RETT cut also creates a fiscal imbalance. Under PE, the government loses 241 euros per household, primarily because of foregone RETT revenue (partially offset by a minor reduction in social housing subsidy). To restore fiscal neutrality in full GE, income tax revenue rises by 183 euros per household (Table 5), which slows asset accumulation. The effect is mild, pushing homeownership only 0.5 pp below the price-adjust specification.²⁹ The fiscal burden falls disproportionately on higher-income households through the progressive tax structure, contributing to the slightly negative demand response in quintiles (Q) Q4 and Q5. The portfolio, life-cycle and social housing patterns observed under PE carry over to GE with smaller magnitudes (Figures 1, 2; Table 5).

²⁷He documents a proportionally similar positive homeownership response to a revenue neutral abolition of a 2% property transfer tax.

²⁸The implied semielasticity of about -0.5% per percentage point is smaller than the short run estimate of approximately -3% reported by Dolls et al. (2025) for Germany.

²⁹The mild effect of budget neutrality is also visible in the similarity between the PE and tax-adjust specifications (Tables 6 and A2).

Table 5: Counterfactual results: General equilibrium (prices and taxes adjust).

	Benchmark	CI1	CI2	CI3	CI4
Homeownership (%)	42.5	49.7	47.2	46.1	53.2
– 25–34 years	13.2	18.6	14.4	14.3	18.6
– 35–44 years	33.0	41.7	35.2	35.2	43.3
– 45–54 years	52.7	64.8	55.4	55.4	66.9
– 55–64 years	61.2	73.2	64.4	64.5	75.9
– 65+ years	47.4	50.0	56.8	53.4	57.2
Total wealth	128.7	137.5	132.4	131.6	138.9
– Housing	85.7	104.0	86.8	87.9	104.7
– Financial	46.7	39.1	48.6	47.1	39.4
– Mortgage	–3.6	–5.5	–3.0	–3.4	–5.2
House price p	1.000	1.013	0.978	0.987	0.995
Price-to-rent ratio	18.38	18.46	18.23	18.29	18.34
Δ Gov't BC (per HH)	—	0.000	0.000	–0.000	–0.000
– Δ RETT revenue	—	–0.194	–0.026	–0.009	–0.197
– Δ Inc. tax revenue	—	0.183	0.124	0.054	0.231
– Δ SH subsidy	—	–0.012	–0.070	–0.059	–0.061
– Δ RTB cost	—	0.000	0.168	0.104	0.096
Δ Demand (%)	—	0.56	–0.95	–0.59	–0.23
– Income Q1	—	1.49	0.58	0.68	2.13
– Income Q2	—	1.40	–1.15	–0.80	0.46
– Income Q3	—	0.82	–1.48	–1.17	–0.61
– Income Q4	—	–0.18	–1.13	–0.81	–1.19
– Income Q5	—	–0.00	–0.99	–0.45	–0.73
SH stock share (%)	7.11	5.96	1.39	1.06	0.98
RTB homeowners (%)	0.00	0.00	8.94	5.43	5.01
Gap closed (%)	—	27.6	17.9	13.6	40.9

Note: Monetary values in thousands of 2006 euros. HH stands for Household, while Q represent the different income deciles. Gap closed is $(\text{HOR}_{\text{CI}} - 42.5) / (68.7 - 42.5) \times 100$, where 68.7% is the Italian homeownership rate from pooled SHIW 2010 and 2014 (see Appendix G).

Quantitatively, the Italian RETT cut produces an homeownership gain in general equilibrium of 7.2 pp, compared with the 8.2 pp obtained by Kaas et al. (2021) for the US.³⁰ The marginal homeownership gain per percentage point of RETT reduction is 1.86 pp, corresponding to 9.81% more transactions per percentage point and an average transaction elasticity of 0.27. Both estimates fall slightly above the empirical range reported by Fritzsche and Vandrei (2019) (approximately 7%) and Petkova and Weichenrieder (2017) (elasticity of 0.23).³¹

6.2 CI2

Implementing the right-to-buy mechanism within the German social housing system raises homeownership to 47.2% (+4.7 pp) in general equilibrium (Table 5), closing 17.9% of the gap. Unlike CI1, where the homeownership gain operates through the rent-versus-buy margin of private-market households, the RTB channel works through the direct conversion of social housing tenants into locked-in homeowners.

Under PE, homeownership rises to only 45.1% (+2.6 pp; Table 6), also because the RTB

³⁰Notice that Kaas et al. (2021) keep the social housing stock fixed at the benchmark value, while this thesis does not. Thus, these two numbers cannot be directly compared.

³¹As Kaas et al. (2021) notes, this might be due to the pool of housing units considered in the calculation.

Table 6: Counterfactual results: Partial equilibrium (fixed prices, fixed taxes).

	Benchmark	CI1	CI2	CI3	CI4
Homeownership (%)	42.5	51.2	45.1	45.0	52.5
– 25–34 years	13.2	18.9	13.8	13.9	18.5
– 35–44 years	33.0	42.8	33.5	34.4	42.4
– 45–54 years	52.7	65.3	53.2	54.2	66.1
– 55–64 years	61.2	74.1	62.4	63.1	75.0
– 65+ years	47.4	53.1	53.7	52.2	56.6
Total wealth	128.7	140.1	130.7	130.7	139.6
– Housing	85.7	106.7	83.7	86.2	104.4
– Financial	46.7	39.0	49.9	47.8	40.2
– Mortgage	–3.6	–5.6	–2.9	–3.3	–5.1
House price p	1.000	1.000	1.000	1.000	1.000
Price-to-rent ratio	18.38	18.38	18.38	18.38	18.38
Δ Gov't BC (per HH)	—	–0.241	–0.111	–0.050	–0.290
– Δ RETT revenue	—	–0.191	–0.037	–0.015	–0.197
– Δ Inc. tax revenue	—	–0.067	0.031	0.011	–0.057
– Δ SH subsidy	—	–0.017	–0.069	–0.058	–0.061
– Δ RTB cost	—	0.000	0.173	0.105	0.097
Δ Demand (%)	—	2.28	–2.33	–1.50	0.17
– Income Q1	—	3.40	–1.11	–0.41	2.39
– Income Q2	—	3.04	–3.00	–1.83	0.65
– Income Q3	—	2.37	–2.80	–2.05	–0.08
– Income Q4	—	1.53	–2.32	–1.59	–0.66
– Income Q5	—	1.78	–2.15	–1.31	–0.35
SH stock share (%)	7.11	5.77	1.42	1.06	0.98
RTB homeowners (%)	0.00	0.00	9.07	5.47	5.05
Gap closed (%)	—	33.3	9.8	9.6	38.2

Note: Monetary values in thousands of 2006 euros. HH stands for Household, while Q represent the different income deciles. Gap closed is $(\text{HOR}_{\text{CI}} - 42.5) / (68.7 - 42.5) \times 100$, where 68.7% is the Italian homeownership rate from pooled SHIW 2010 and 2014 (see Appendix G).

produces a contraction in aggregate housing demand (–2.33%) that cannot translate into lower prices. The contraction reflects the flow-versus-stock asymmetry described in Section 5.2: even at a 68% discount, the typical purchased unit is roughly 27% smaller than the services previously consumed as a tenant (Table A5). When prices are allowed to adjust, the house price falls to 0.978 and the price-to-rent ratio declines to 18.24 (Table A1), making private-market purchase cheaper and increasing homeownership gains by 1.5 percentage points. This is the opposite of the CI1 pattern: whereas the RETT reduction generates demand that pushes prices up, the RTB generates a supply surplus that pulls prices down.

The social housing stock falls from 7.11% to around 1.4% (Table 5, 6), driven by the RTB take ups without compensating construction. Notice that the depletion is self-limiting, since a smaller stock exposes fewer tenants to future RTB offers, lowering the volume of tenure conversions in steady state. The composition of the remaining sector shifts visibly, as shown in general equilibrium by Table A4: older tenants have spent more periods exposed to RTB offers and are cumulatively more likely to have exercised, so the retiree share collapses from 30.1% to 7.4% while the 25–34 share more than doubles. Along the wealth dimension, wealthier tenants are more likely to meet the self-financing constraint and exit, leaving the bottom three deciles as a larger fraction of the residual tenant pool. The sector undergoes an endogenous

residualisation, converging toward a system that serves younger and poorer households, a pattern that foreshadows the design of the Italian system examined in CI3. The stock depletion has also a fiscal counterpart: the reduced construction subsidy partially offsets the cost of the RTB discount, leaving a net deficit of 111 euros per household in PE, restored in general equilibrium by an income tax increase of 124 euros, a smaller fiscal adjustment than CI1 requires (Tables 5, 6).

While the aggregate homeownership gain may appear modest relative to the attractiveness of the RTB option, the decomposition in Table 7 reveals two offsetting effects. The mechanism converts 8.9% of the population into locked-in homeowners, but private-market homeownership simultaneously falls by 4.2 pp, yielding a net gain of 4.7 pp, consistent with the crowding-out predictions of C. Hilber (2007). Two channels likely drive the decline. Households who in the benchmark would have accumulated wealth in social housing and eventually purchased on the private market now exercise the RTB option instead, since the discounted purchase is substantially cheaper. At the same time, the depleted social housing stock leaves more households in the private rental market without the wealth to purchase at market prices. The equilibrium price decline attenuates this crowding-out but does not eliminate it. The age pattern in Table 7 shows that crowding-out is strongest for 45–64 year olds, where private-market homeownership falls by 7–8 pp against 10–12 pp of RTB conversions. Among retirees, by contrast, the private-market decline is negligible, so the full RTB conversion translates into a homeownership gain.

The life-cycle and wealth profiles (Figures 1, 2, top right) reflect this decomposition directly. Gains are modest at working ages (+1 to +3 pp) and largest for retirees (+9.4 pp). By wealth decile, the gains appear concentrated in deciles (D) D4–D6, but this is partly a compositional artifact. The wealth transition matrix (Table A6) shows that almost no RTB exerciser remains in the pre-exercise decile: exercisers originate primarily from D3 (36.8% of mass) and D4 (21.4%) but land in D5–D6 once the purchased house enters the balance sheet at market value, producing an average (though illiquid) wealth jump. Evaluating locked-in owners at their pre-exercise position shifts the gains into the lower deciles (Figure A1), showing that the mechanism reaches households that would not have become owners through the private market.

On average, in general equilibrium, 76.3% of tenants who receive the purchase opportunity exercise it (Table A3). Take-up depends heavily on asset accumulation, surging from 22.5% in D1 to over 94% from D3 onward. This arises because the strict self-financing constraint prevents the poorest households from affording even the discounted purchase price. Take-up also exhibits a pronounced life-cycle profile: it rises from 65.2% at ages 25–34 to a peak of 88.2% at 55–64, before dropping sharply to 55.3% for retirees. Young households exercise the option less frequently because they lack sufficient savings and face higher utility costs from the permanent resale ban, given their long horizon of potential housing size adjustments. Peak take-up occurs at 55–64 when households have reached their peak wealth and their housing needs have stabilized. Finally, the drop among retirees reflects the asset-decumulation phase of the life cycle: elderly households have a shorter remaining lifespan to amortize the large, illiquid upfront cost of the unit through future rent savings.

Section 5.2 identified two competing effects of the RTB option on tenant saving: a wealth

Table 7: Homeownership Changes: RTB conversions vs. private-market changes (GE).

Age	CI2			CI3			CI4		
	Δ HOR	RTB	Priv.	Δ HOR	RTB	Priv.	Δ HOR	RTB	Priv.
25–34	+1.2	+3.4	–2.2	+1.1	+2.0	–0.9	+5.4	+1.9	+3.5
35–44	+2.2	+7.2	–5.0	+2.1	+4.4	–2.2	+10.2	+4.2	+6.1
45–54	+2.7	+9.7	–7.0	+2.8	+5.9	–3.2	+14.2	+5.5	+8.7
55–64	+3.2	+11.7	–8.4	+3.3	+7.1	–3.8	+14.7	+6.5	+8.3
65+	+9.4	+10.9	–1.4	+6.0	+6.6	–0.6	+9.8	+6.0	+3.9
All	+4.7	+8.9	–4.2	+3.6	+5.4	–1.9	+10.7	+5.0	+5.7

Note: RTB is the population share in the locked-in state. Priv. is the change in the fraction of private market households households with $h > 0$ relative to the benchmark. The two components sum to Δ HOR. CI1 is omitted since it has no RTB mechanism: at the aggregate level Δ HOR = Priv. = +7.2 pp.

accumulation motive and an insurance effect. Figure A4 reports the change in the savings decision rule at the benchmark distribution for social housing tenants that have not yet received a RTB offer under PE, isolating the direct effect on savings behavior. The accumulation motive dominates at every age, and is strongest for the 35 to 44 group, whose remaining tenancy is long enough for the option to be appearing but whose wealth is still low enough that additional saving affects whether they will be able to exercise. The magnitude is modest but confirms that tenants respond to the anticipated opportunity by building a larger financial buffer.

Three features of this specification produce a conservative estimate of the RTB contribution. The permanent lock-in depresses the value of exercising the option relative to the actual finite restriction. Zero replacement of purchased units contracts the social housing sector, reducing the pool of tenants exposed to future offers. The self-financing constraint excludes the poorest tenants, so the households most in need of the implicit wealth transfer are the least able to access it. The measured gain of 4.7 pp should therefore be read as a lower bound on the potential contribution of the RTB mechanism.³²

6.3 CI3

CI3 implements the three defining features of the Italian social housing design simultaneously: restricted eligibility, a deeper rent discount, and the RTB mechanism. Homeownership rises to 46.1% (+3.6 pp) in general equilibrium (Table 5), closing 13.6% of the gap. Relative to CI2, the two additional Italian features shrink rather than amplify the aggregate contribution, because they simultaneously reduce the eligible pool and raise the value of remaining a tenant.

Under PE, homeownership rises to 45.0% (+2.5 pp; Table 6), nearly identical to CI2's PE result (+2.6 pp), while the demand contraction is smaller than under CI2 (given fewer RTB conversions, Table 7). When prices adjust, the house price falls to 0.988 and the price-to-rent ratio declines to 18.30 (Table A1), making owning relatively more attractive than renting. The fiscal imbalance is the smallest of the four counterfactuals: the RTB discount cost (105 euros per

³²For comparison, the UK right-to-buy is estimated to have raised aggregate homeownership by roughly 15 percentage points between 1979 and the early 2000s (Disney and Luo 2017), although the British policy operated under a more permissive regime (finite resale restriction, mortgage finance allowed against the discounted unit, and the much larger pre reform council housing stock).

household) is partially offset by the reduced construction subsidy (−58 euros), yielding a net PE deficit of 50 euros (Table 6). In GE, budget balance is restored by an income tax increase of 54 euros, leaving the homeownership result close to the price-adjust specification.

The social housing stock falls further than under CI2, to 1.06% in GE, since restricted eligibility limits inflows on top of the RTB induced reduction. The composition shift is also sharper (Table A4): the bottom three wealth deciles account for 68.1% of the tenant population and the retiree share edges above the CI2 level (8.8% vs 7.4%) as the deeper discount gives some elderly tenants a stronger incentive to remain. This effectively completes the residualisation process started under CI2: restricted targeting ensures that only the bottom three income deciles enter, and the RTB channel removes those who accumulate enough resources to exercise.

The HOR decomposition (Table 7) shows that the smaller aggregate gain reproduces the CI2 pattern in attenuated form. RTB conversions account for +5.4 pp (from CI2's +8.9 pp) and the private-market decline is −1.9 pp (against CI2's −4.2 pp). The smaller RTB contribution reflects two compounding forces: the restricted pool exposes fewer tenants to offers, and the deeper rent discount lowers take-up conditional on eligibility (average 63.9% versus 76.3% in CI2, Table A3). The smaller private-market decline arises since fewer of the households who exercised the option could have purchased on the private market. By age (Table 7), crowding-out is roughly halved relative to CI2 at every working age, and is negligible for retirees.

The anticipation savings effect is stronger under CI3 than under CI2 (Figure A4), with the largest difference at 35–44 age bracket. The deeper rent discount frees disposable income, which leaves tenants more room to accumulate. The diagnostic cannot, however, distinguish a genuinely stronger anticipation motive from the mechanical effect of higher disposable income, so the comparison should be read as an upper bound on the additional anticipation response.

The life-cycle profile of homeownership gains follows the CI2 pattern in smaller magnitude, with gains concentrated among retirees (+6.0 pp) and the 55–64 bracket (+3.3 pp) thanks to the cumulative stock of past RTB exercises (Figure 1, bottom left). Along the wealth distribution, major gains appear between D4 and D6, but as in CI2 this partly reflects the wealth jump induced by RTB exercise: evaluating locked-in owners at their pre-exercise position shifts the gains into the bottom deciles (Figure A2). The housing demand response by income quintile in GE (Table 5) shows that Q1 demand rises (+0.68%) as the poorest households benefit from the deeper rent discount. Conversely, demand falls across Q2–Q5. This decline occurs because, on one side, households exercising the RTB option purchase smaller units than they previously rented, while those excluded from the social housing system now face higher rents.

RTB take-up rates decline compared to CI2 across all age groups and wealth deciles (Table A3). The drop is driven primarily by young households, for whom the deeper rent discount makes social housing tenancy a more valuable insurance arrangement given the longer horizon ahead. At the bottom of the wealth distribution, D1 take-up collapses to 6.4%, since the combination of deeper rents and a tight borrowing constraint makes the option unattractive.

Taken together, CI3 closes less of the homeownership gap than CI2 (13.6% versus 17.9%) but represents a qualitatively different intervention. The German eligibility rules in CI2 expose a

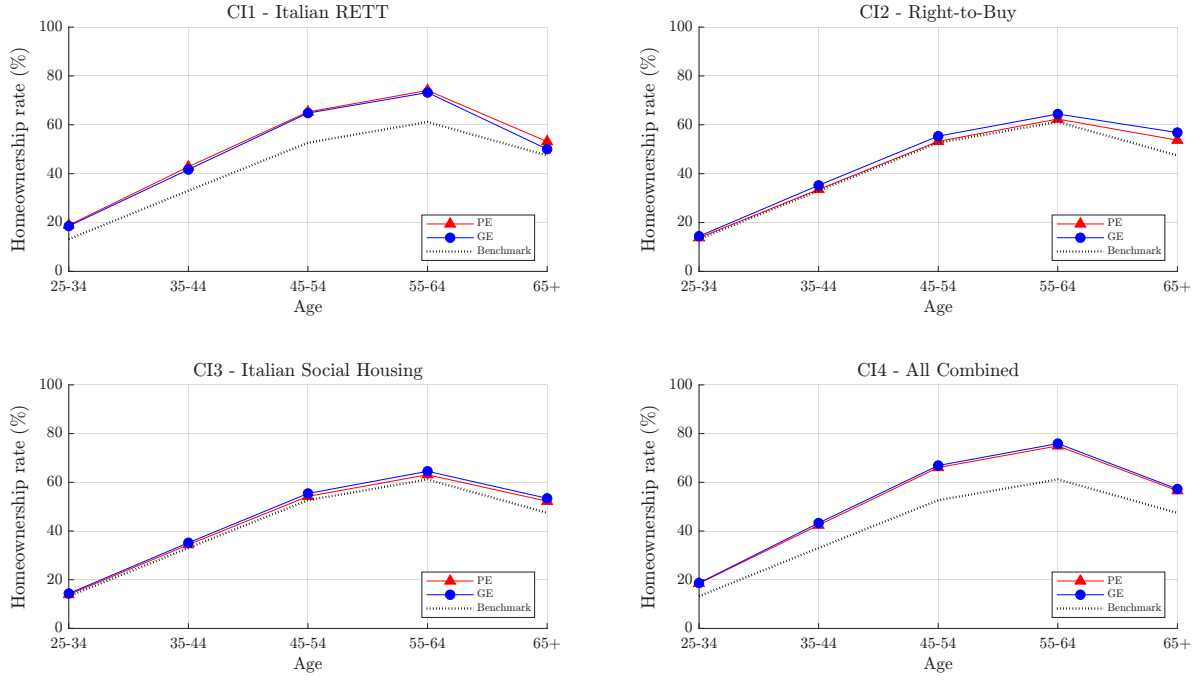


Figure 1: Homeownership rate by age group: Italian counterfactuals vs. benchmark.

larger pool to the RTB option, producing a larger homeownership gain at a higher fiscal cost; the Italian design restricts access and yields lower take-up, but provides a more generous safety net for those who remain at a smaller cost. The +3.6 pp aggregate gain is comparable to both the effect of abolishing social housing entirely in Kaas et al. (2021) (+4.0 pp) and to their targeted housing subsidy experiment (+3.6 pp). The conservative features noted for CI2 apply here.

6.4 CI4

Combining the lower RETT with the full Italian social housing system raises homeownership to 53.2% (+10.7 pp) in general equilibrium (Table 5), closing 40.9% of the Germany–Italy gap. The combined effect is close to the sum of CI1 (+7.2 pp) and CI3 (+3.6 pp), because the two channels reach largely distinct segments of the population: the RETT reduction shifts the rent-versus-buy margin for private-market households, while the Italian social housing design converts tenants into owners through the RTB option.

The HOR decomposition in Table 7 clarifies the interaction. RTB conversions contribute +5.0 pp and the private-market component contributes +5.7 pp. The key difference from CI2 and CI3 is the sign of the private-market component: in those experiments, the RTB diverts households who would otherwise have purchased on the private market; in CI4, the simultaneous RETT reduction more than compensates for this diversion, leaving the private-market component positive at every age (Table 7, Panel B). The combination thus neutralizes the crowding-out mechanism documented in Section 6.2.

The equilibrium house price behaves similarly. CI1 in isolation pushes the price up to 1.013, and CI3 in isolation pushes it down to 0.987. When combined, these opposing forces nearly cancel, leaving the CI4 price at 0.995 and the price-to-rent ratio essentially unchanged. The PE-GE

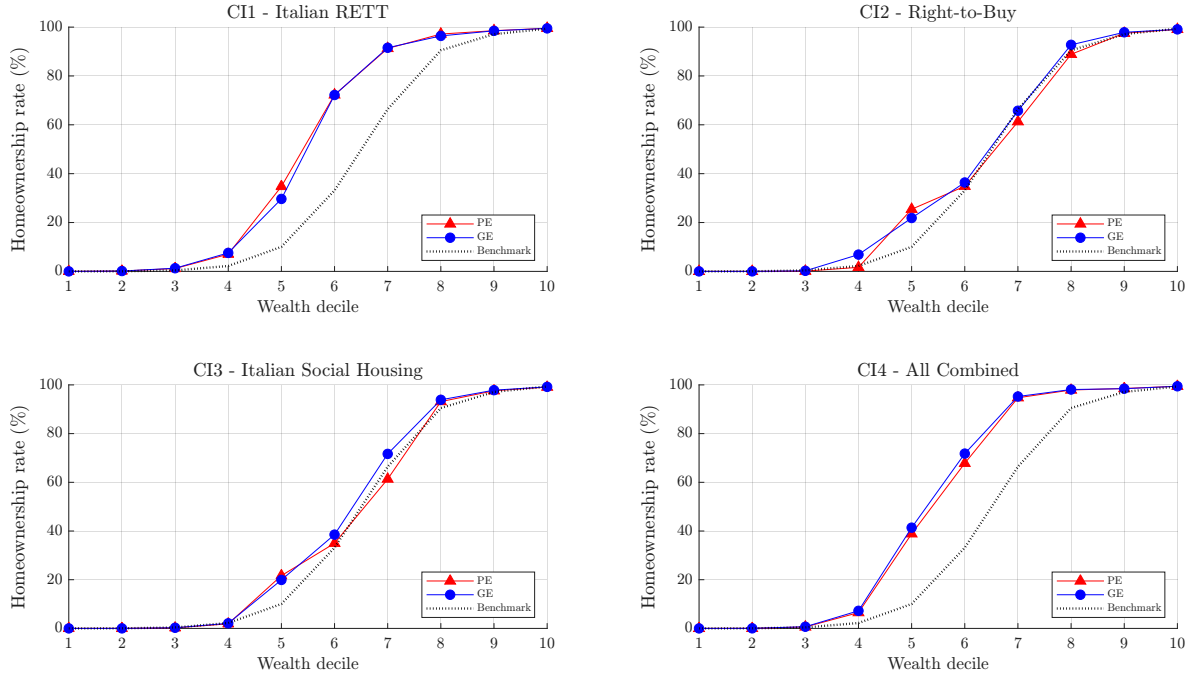


Figure 2: Homeownership rate by wealth decile (working age): Italian counterfactuals vs. benchmark.

wedge is correspondingly small, with both specifications delivering similar homeownership rates.

The life-cycle profile (Figure 1, bottom right) reflects the superposition of the two channels. The large gains at 45–54 and 55–64 (+14.2 and +14.7 pp) are driven primarily by the RETT channel, which shifts the rent-buy margin for middle-aged households with sufficient assets. The retiree gain (+9.8 pp) inherits from the cumulative stock of past RTB exercises. By wealth decile (Figure 2, bottom right), gains are concentrated between D4 and D8, and as in CI3 the pre-RTB figure (Figure A3) confirms that RTB exercisers originate from the bottom deciles and are reclassified upward upon acquiring housing wealth.

CI4 produces the largest portfolio shift of the four counterfactuals (Table 5): housing wealth rises to 104.7 thousand euros, financial assets fall to 39.4, and mortgage debt grows to –5.2. The RETT channel drives the substitution from financial to housing wealth through the mechanism identified in Section 6.1, while the RTB channel adds illiquid housing to locked-in owners who would otherwise have struggled to purchase. The net effect is a population with higher total wealth (138.9 versus 128.7 thousand euros) but a substantially less liquid portfolio.

In general equilibrium, the social housing stock falls to 0.98%, with 5.01% of the population in the locked-in state. Take-up is 63.2%, slightly below CI3 (63.9%, Table A3), since the lower RETT makes private-market purchase cheaper and induces a small fraction of tenants who would have exercised RTB under CI3 to exit through the private market instead. On the fiscal side, the RETT revenue loss (–197 euros per household) is the dominant cost, but is offset in GE by higher income tax revenue (+231 euros) and reduced social housing expenditure (–61 euros).

6.5 Discussion

The above analysis suggests that differences in RETT and social housing systems close approximately 41% of the Germany–Italy homeownership gap. Understanding where the remaining gap concentrates helps identify which factors the present framework does not capture.

Figure 3 compares the life-cycle homeownership profiles in the benchmark, the CI4 experiment, and the pooled 2010 and 2014 waves of the SHIW. Implementing the Italian housing policies nearly eliminates the gap for middle-aged households: the residual for the 45–54 and 55–64 groups is only 1.8 and 1.6 pp respectively. Young (25–34) and elderly (65+) households, instead, still exhibit substantially lower rates than the Italian level, with an intermediate gap for 35–44 year olds. Along the wealth distribution (Figure 4), the gap is essentially closed from D7 onward but remains sizable in the middle deciles, though substantially reduced relative to the benchmark.³³ Figure A5 combines the two dimensions and confirms that part of the gap is concentrated among young, middle-wealth households: the 45–54 and 55–64 profiles track the Italian data from D5 onward reasonably well, while the 25–44 group lags in the middle of the wealth distribution.

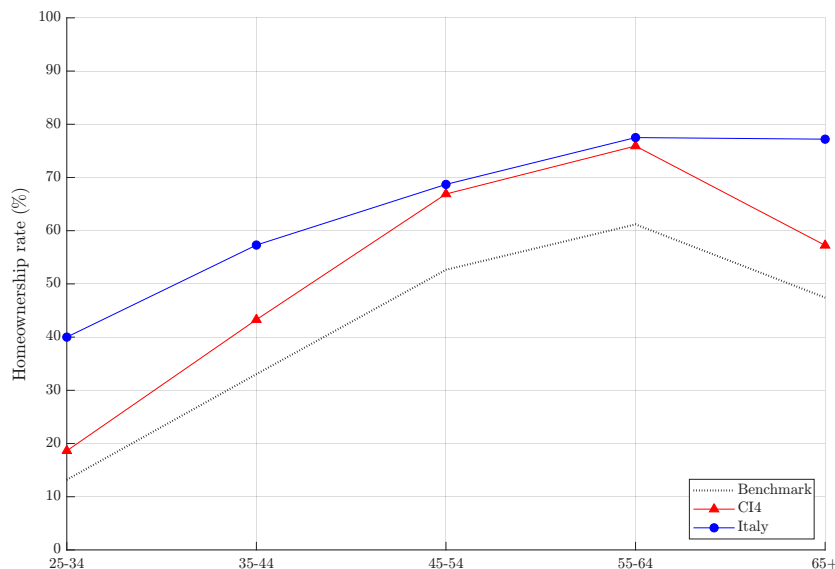


Figure 3: Homeownership rate by age group: Germany vs Italy vs CI4 (GE).

Part of the residual for young low-wealth households may reflect known limitations of the benchmark model’s fit to German data (see Section 3) and the conservative RTB specification. The magnitude of the cross-country gap in some deciles, however, exceeds what these modeling choices could account for, suggesting the presence of omitted factors.

Italian young adults co-reside with their parents at more than twice the German rate (Grevenbrock, Ludwig, and Siassi 2024), providing a zero-cost housing arrangement that facilitates down-payment accumulation without appearing as an independent renter household. Taken this factor into account, Grevenbrock, Ludwig, and Siassi (2024) show that individual-level homeownership rates are virtually identical between Germany and Italy until age 48, suggesting

³³Wealth deciles are computed within each economy’s own working-age wealth distribution (for Italy using pooled SHIW 2010 and 2014; for the models using the experiment-specific stationary distribution), so that each decile label represents the same relative position rather than the same nominal wealth level.

that part of the household-level gap for young adults reflects differences in household formation timing rather than tenure choice. Guiso and Tullio (2002) also show that about one-third of Italian

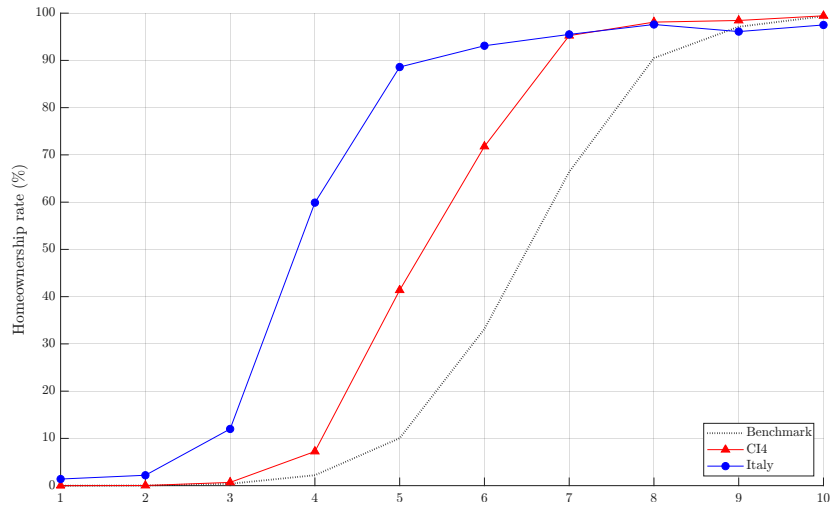


Figure 4: Homeownership rate by wealth decile (working-age): Germany vs Italy vs CI4 (GE).

homeowners received their dwelling as a gift or bequest, or benefited from financial support for its purchase. This channel is relevant for both private-market ownership and RTB units (with some limitations), but is not captured in this model with only accidental bequests upon death. Furthermore, Italian wages increase considerably later in life than German ones, thus making parental support even more relevant at the first stages of the life cycle (Grevenbrock, Ludwig, and Siassi 2024). Pay-as-you-go pensions are also generally more generous (OECD 2025), reducing the incentive to decumulate housing wealth in retirement and potentially explaining part of the residual for the 65+ group (Nakajima and Telyukova 2020). Fehr, Hofmann, and Kudrna (2025) show, however, that less generous pensions also raise savings during working life, reducing housing wealth into retirement. Which of these two opposing channels dominates is not clear a priori. Beyond income and transfers, other institutional differences may matter. As noted in Section 3, Germany’s rental market is structurally more secure than the Italian one, with plausible effects on homeownership decisions (Kindermann and Kohls 2018). Italy’s exemption of first-time buyers from the municipal property tax provides an additional wedge in the tenure choice that the present setting does not capture. Finally, homeownership carries a stronger social and familial significance in Italy than in Germany, where renting is a socially accepted long-term arrangement (Huber and T. Schmidt 2019; Giuliano 2007); such preference differences would manifest as a higher effective weight on housing in the utility function.

7 Welfare Analysis

While Section 6 provided a positive understanding of whether a reduction in transfer taxes and a different social housing program can account for the Italy–Germany homeownership gap, it did not address the welfare consequences of these policies, which matter for the normative assessment of the reform. This section therefore evaluates CI1–CI4 under the same four

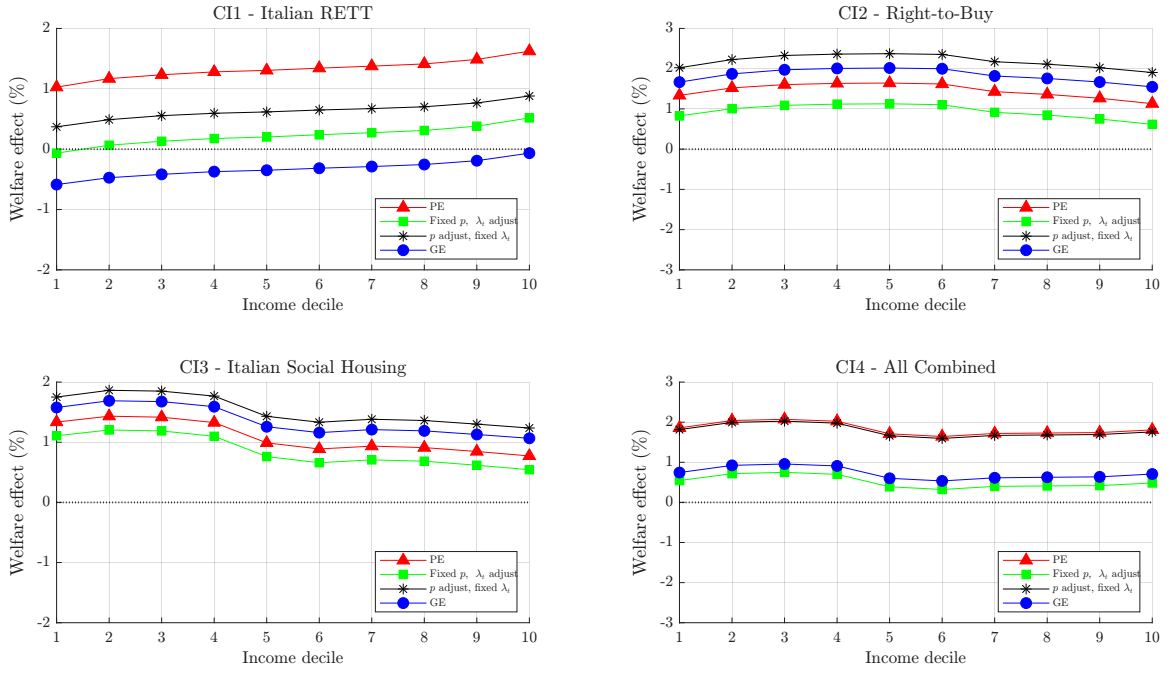


Figure 5: Welfare effects by income decile for counterfactual experiments (consumption equivalent variation).

equilibrium specifications as Section 6. Following Kaas et al. (2021), I evaluate welfare in terms of consumption equivalent variation (CEV) for newborn households entering the stationary equilibrium after drawing their first income realization. The CEV measures the uniform percentage change in consumption that a household born into the benchmark economy would need to receive in every period and state to be indifferent between remaining in the benchmark and being born instead into the counterfactual economy. For a newborn with initial income decile i and zero assets ($a = 0, h = 0$), let $\mathbb{E}[V_0(i)]$ and $\mathbb{E}[V_1(i)]$ denote expected lifetime utility in the benchmark and the counterfactual. Given the Cobb–Douglas/CRRA specification of the period utility (1), $CSV(i) = (\mathbb{E}[V_1(i)]/\mathbb{E}[V_0(i)])^{1/[\zeta(1-\gamma)]} - 1$, with the exponent following from the homogeneity of utility in consumption. A positive CEV indicates the household is better off in the counterfactual. Since the measure contrasts agents spending their entire life in each stationary equilibrium, it does not capture the welfare of generations alive during the transition. The focus on newborns is standard in the quantitative housing literature (Sommer and Sullivan 2018; Nakajima 2020; Kaas et al. 2021) and has the advantage of comparing households at the same initial state. Reporting the CEV by income decile then reveals the distributional consequences of the policy. The results are shown in Figure 5.

Reducing the RETT to the Italian level (CI1) generates a positive PE welfare effect ranging from 1.0 to 1.6% along the income distribution, more pronounced for higher-income households where most marginal house buyers lie and who therefore benefit most from the reduction in transaction costs. These gains dwindle as equilibrium adjustments are accounted for. When house prices adjust, the effects are roughly halved, particularly for lower-income households who are less likely to purchase and more exposed to rent fluctuations. Replacing the lost transfer tax revenue with higher income taxes compresses the gains further, leaving the lowest deciles near

zero or mildly negative. In full GE, the policy produces a negative welfare effect across all deciles (as in Kaas et al. (2021) and D. J. Schmidt (2022)), most pronounced for poorer households facing higher income taxes without the offsetting benefit of cheaper home purchase.

Introducing the RTB mechanism within the German social housing system (CI2) produces positive welfare effects across all income deciles and all specifications. Under PE, gains range from 1.1 to 1.6%, relatively uniform across the distribution, reflecting the broad eligibility of the German system. Three channels drive the gain. Upon exercising, the household pays a one-time discounted price and then consumes housing services, freeing resources for non-housing consumption. The locked-in state removes the risk of losing social housing access and returning to the private rental market, providing insurance in an incomplete-markets environment. Finally, homeowners receive the retirement utility premium ξ_5 that tenants do not. The gains are largest when only house prices adjust, since the price decline benefits both exercisers and non-participants, producing welfare gains of around 2.0%. In GE, the welfare effect remains positive, ranging from 1.5% at D1 to 2.0% at D3 and declining mildly toward the top.

Implementing the Italian social housing design alongside the RTB (CI3) also produces positive welfare effects, with a distribution more clearly tilted toward lower-income households, since those are the ones most directly benefiting from the new policy. Under PE, the first three to four income deciles gain 1.3–1.4%, while the upper deciles gain 0.8–0.9%. As in CI2, price adjustment amplifies the gains while tax adjustment compresses them: in full GE, the first four deciles gain approximately 1.6% and the upper deciles around 1.1%. This profile is at odds with Disney and Luo (2017), where a RTB option reduces aggregate welfare of low-income households when council housing quality is high enough to attract middle-wealth purchasers, and can be positive when quality is low. This is due to the fact that: the equilibrium price decline induced by lower housing demand benefits all households in this model; restricted eligibility further prevents middle-income households from entering social housing, directly addressing the main factor driving the negative result in Disney and Luo (2017); finally, the present model abstracts from housing quality heterogeneity, and the decision to accept the RTB offer depends entirely on the tenant’s financial position and the discount size. This may overstate the welfare gain if Italian units are of lower quality than their market equivalents. The CI3 profile also contrasts with the abolition of social housing in Kaas et al. (2021), where the largest welfare gains accrue to high-income households, while it more closely resembles the positive welfare effects stemming from their targeted housing subsidy experiment on the two lowest income deciles.

Combining these policies (CI4) raises welfare across the entire income distribution in GE. The two channels benefit largely distinct populations, consistent with the homeownership results in Section 6. The bottom four deciles gain 0.7–0.9%, driven primarily by the RTB option, while the top three deciles gain 0.6–0.7%, driven by the RETT reduction that lowers the cost of sizable housing purchases. The price adjustment documented in Section 6.4 shows up in welfare as well: the PE and price-adjust specifications produce nearly identical welfare profiles, confirming that the opposing demand effects of the two channels stabilize the equilibrium price and prevent the welfare losses that the RETT channel generates in isolation. This contrasts with the combined

reform in Kaas et al. (2021), where the full GE welfare effect is negative, driven by the strong price increase from the RETT reduction and mortgage deduction channels.

8 Conclusion

This thesis has examined whether differences in housing policies between Germany and Italy can account for the homeownership gap between the two countries. Taking Kaas et al. (2021)'s overlapping-generations model as a basis, I implemented four counterfactual experiments transplanting distinctive features of the Italian housing system in the German economy: a reduced real-estate transfer tax, a right-to-buy (RTB) mechanism within the German social housing system, a social housing design with restricted eligibility, deeper rent discount and RTB along Italian lines, and the combination of the RETT reduction with the Italian social housing design.

The RETT reduction alone accounts for almost one third of the gap and constitutes the single most important lever, operating through the rent-versus-buy margin of middle-aged, middle-wealth households. The right-to-buy mechanism, in the conservative specification adopted in this thesis, closes 17.9% of the gap, with its direct conversion channel partially offset by a crowding-out of private-market ownership. The Italian social housing design produces a smaller aggregate effect than the RTB alone, as the restricted targeting reduces the pool of tenants exposed to (and capable to profit of) the purchase opportunity and the deeper rent discount raises the value of remaining a tenant. Combining the reduced RETT with the Italian social housing design closes roughly 41% of the homeownership gap, but unevenly: while there are almost no differences among middle aged workers, gaps still persist among young middle wealth workers and retirees. A central finding is also the different role played by the Italian social housing system compared to the German one. Restricted eligibility and the deeper rent discount generate an residualisation of the sector, that serves the poorest renters while opening a path to homeownership for otherwise excluded households through the RTB option.

The welfare effects diverge across policies. The RETT reduction in isolation produces positive partial equilibrium gains that turn mildly negative once the rise in house prices and the compensating income-tax increase are taken into account (especially for poorer households). Both the right-to-buy mechanism and the Italian social housing design produce positive welfare effects across the income distribution, with gains tilted toward lower-income households, and the combined policies deliver positive welfare gains across all deciles.

Notwithstanding these findings, a portion of the Germany–Italy gap remains unexplained. Household formation differences, inter-generational transfers, income trajectories, pension generosity, rental market regulation, and property tax design provide additional channels that likely interact with the housing-policy counterfactuals analyzed here, while cultural differences in the valuation of homeownership relative to renting may amplify the tenure-choice response to any given policy configuration. A less restrictive modeling of the RTB resale regime would also bring the specification closer to its institutional reality. Several directions for future work follow naturally from these considerations.

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A Tables and Graphs

This appendix section presents two summary tables of the counterfactual experiments under the remaining two equilibrium specifications and diagnostic graphs whose results have been mentioned in the thesis.

Table A1: Counterfactual results: Prices adjust, fixed taxes.

	Benchmark	CI1	CI2	CI3	CI4
Homeownership (%)	42.5	49.2	46.7	46.0	52.4
– 25–34 years	13.2	18.3	14.3	14.3	18.3
– 35–44 years	33.0	41.4	35.1	35.1	42.3
– 45–54 years	52.7	64.2	55.2	55.4	66.0
– 55–64 years	61.2	72.5	64.1	64.4	74.9
– 65+ years	47.4	49.4	55.8	53.3	56.5
Total wealth	128.7	138.2	132.4	131.7	139.5
– Housing	85.7	104.0	86.5	87.9	104.3
– Financial	46.7	39.7	48.9	47.2	40.3
– Mortgage	–3.6	–5.5	–3.0	–3.4	–5.1
House price p	1.000	1.019	0.980	0.988	1.001
Price-to-rent ratio	18.38	18.50	18.24	18.30	18.38
Δ Gov't BC (per HH)	—	–0.247	–0.103	–0.048	–0.290
– Δ RETT revenue	—	–0.194	–0.027	–0.009	–0.197
– Δ Inc. tax revenue	—	–0.062	0.023	0.006	–0.056
– Δ SH subsidy	—	–0.009	–0.070	–0.059	–0.061
– Δ RTB cost	—	0.000	0.169	0.104	0.097
Δ Demand (%)	—	0.85	–0.87	–0.54	0.07
– Income Q1	—	1.80	0.63	0.69	2.30
– Income Q2	—	1.59	–1.34	–0.77	0.57
– Income Q3	—	1.21	–1.33	–1.14	–0.21
– Income Q4	—	0.11	–1.01	–0.70	–0.78
– Income Q5	—	0.24	–0.81	–0.40	–0.42
SH stock share (%)	7.11	6.04	1.39	1.06	0.98
RTB homeowners (%)	0.00	0.00	8.96	5.43	5.06
Gap closed (%)	—	25.6	16.2	13.3	37.9

Note: Monetary values in thousands of 2006 euros. HH stands for Household, while Q represent the different income deciles. The income tax schedule is held fixed at benchmark values; house prices adjust to clear the housing market. The government budget is not balanced. Gap closed is $(HOR_{CI} - 42.5)/(68.7 - 42.5) \times 100$, where 68.7% is the Italian homeownership rate from pooled SHIW 2010 and 2014 (see Appendix G).

Table A2: Counterfactual results: Fixed prices, taxes adjust.

	Benchmark	CI1	CI2	CI3	CI4
Homeownership (%)	42.5	51.3	45.0	45.1	52.6
– 25–34 years	13.2	19.2	13.8	13.9	18.4
– 35–44 years	33.0	43.1	33.8	34.3	42.8
– 45–54 years	52.7	65.4	53.1	54.4	66.2
– 55–64 years	61.2	74.2	62.3	63.3	75.4
– 65+ years	47.4	53.1	53.6	52.3	56.5
Total wealth	128.7	139.1	130.2	130.5	138.4
– Housing	85.7	106.2	83.4	86.2	103.8
– Financial	46.7	38.5	49.6	47.7	39.7
– Mortgage	–3.6	–5.6	–2.9	–3.3	–5.1
House price p	1.000	1.000	1.000	1.000	1.000
Price-to-rent ratio	18.38	18.38	18.38	18.38	18.38
Δ Gov't BC (per HH)	—	0.000	–0.000	0.000	0.000
– Δ RETT revenue	—	–0.192	–0.037	–0.015	–0.198
– Δ Inc. tax revenue	—	0.175	0.141	0.062	0.233
– Δ SH subsidy	—	–0.017	–0.069	–0.058	–0.061
– Δ RTB cost	—	0.000	0.173	0.105	0.096
Δ Demand (%)	—	1.58	–2.71	–1.67	–0.63
– Income Q1	—	2.75	–1.47	–0.57	1.69
– Income Q2	—	2.53	–3.28	–1.91	–0.00
– Income Q3	—	1.66	–3.21	–2.29	–0.99
– Income Q4	—	0.75	–2.69	–1.79	–1.56
– Income Q5	—	1.02	–2.57	–1.46	–1.12
SH stock share (%)	7.11	5.77	1.42	1.06	0.98
RTB homeowners (%)	0.00	0.00	9.06	5.46	5.03
Gap closed (%)	—	33.8	9.7	9.9	38.8

Note: Monetary values in thousands of 2006 euros. HH stands for Household, while Q represent the different income deciles. House prices are held fixed at benchmark values; the income tax parameter λ_τ is scaled to balance the government budget. Gap closed is $(\text{HOR}_{\text{CI}} - 42.5) / (68.7 - 42.5) \times 100$, where 68.7% is the Italian homeownership rate from pooled SHIW 2010 and 2014 (see Appendix G).

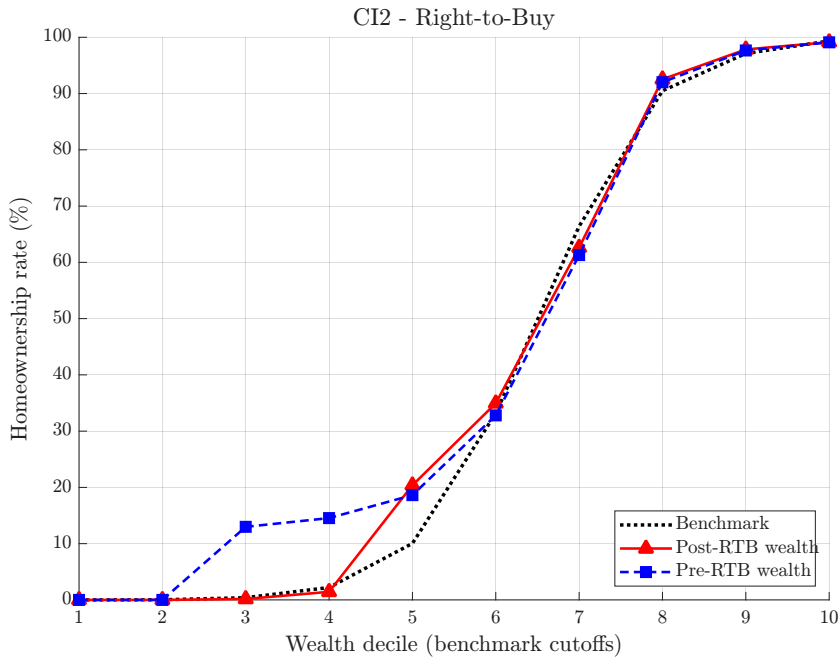


Figure A1: Homeownership rate by wealth decile (working age), Benchmark vs Pre and Post - RTB Wealth: CI2 (GE)

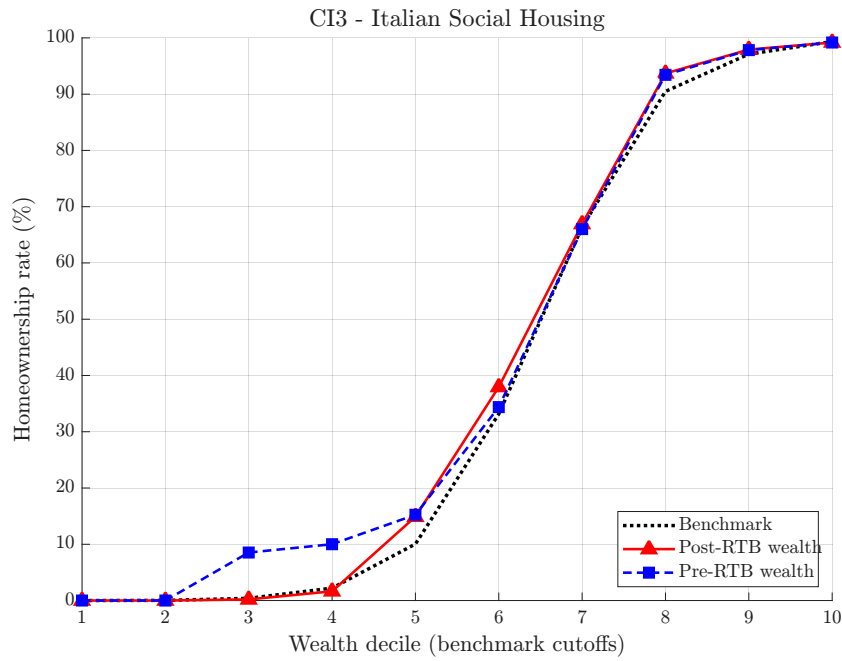


Figure A2: Homeownership rate by wealth decile (working age), Benchmark vs Pre and Post - RTB Wealth: CI3 (GE)

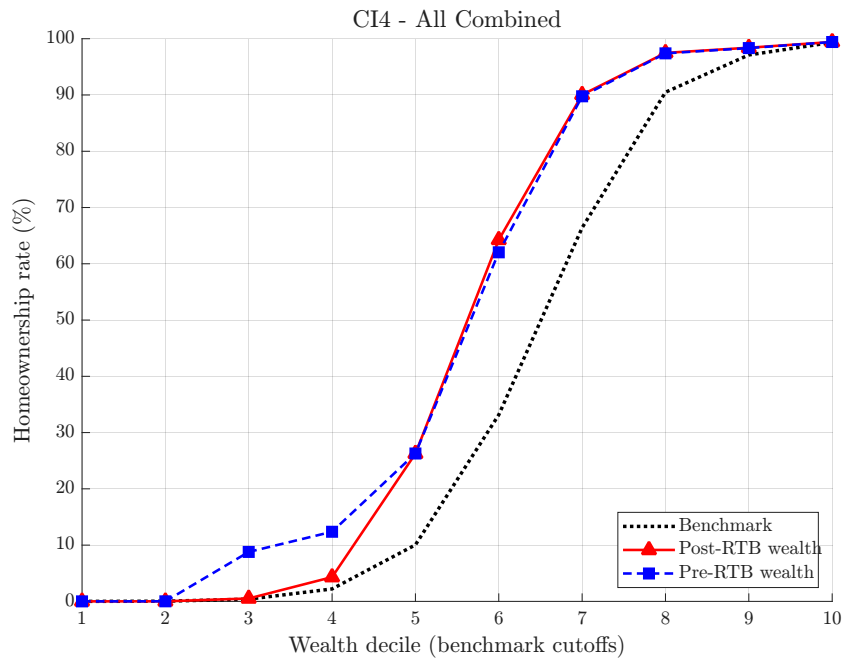


Figure A3: Homeownership rate by wealth decile (working age), Benchmark vs Pre and Post - RTB Wealth: CI4 (GE)

Figure A1, A2 and A3 compare the homeownership rate by wealth decile under two alternative assignments of locked-in owners to positions along the benchmark wealth distribution, in order to ensure comparability across the graphs. The post-RTB curve places them at their current post-purchase wealth. The pre-RTB curve places them at the wealth they were supposed to hold before exercising, reconstructed by adding the purchase cost of the house back to their current financial assets. This reconstruction is exact for households that have just exercised RTB.

However, for households that exercised in earlier periods and have changed their savings since, the reconstructed pre-exercise wealth is not as precise.

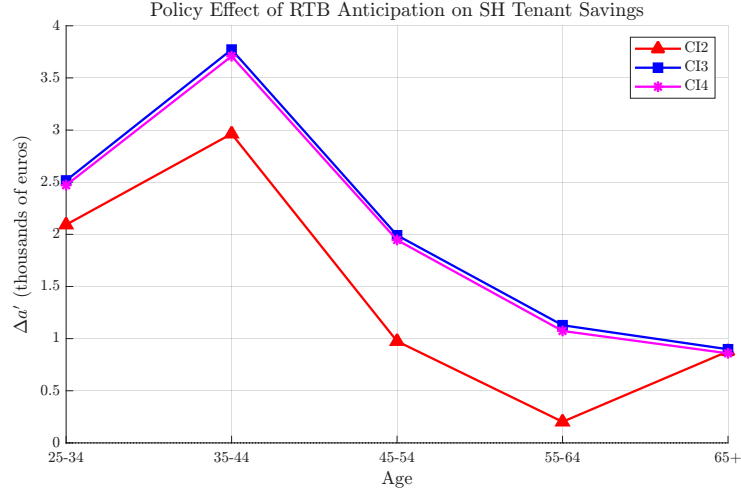


Figure A4: Policy effect of RTB anticipation on social housing tenant savings (PE).

The figure reports $\mathbb{E}_{\text{bench}}[a'_{\text{CF}} - a'_{\text{bench}}]$, i.e. the change in the savings decision rule, at each state, of social housing tenants that have not yet received an RTB offer, weighted by the benchmark distribution under PE. Positive values indicate that the wealth accumulation motive dominates the insurance effect.

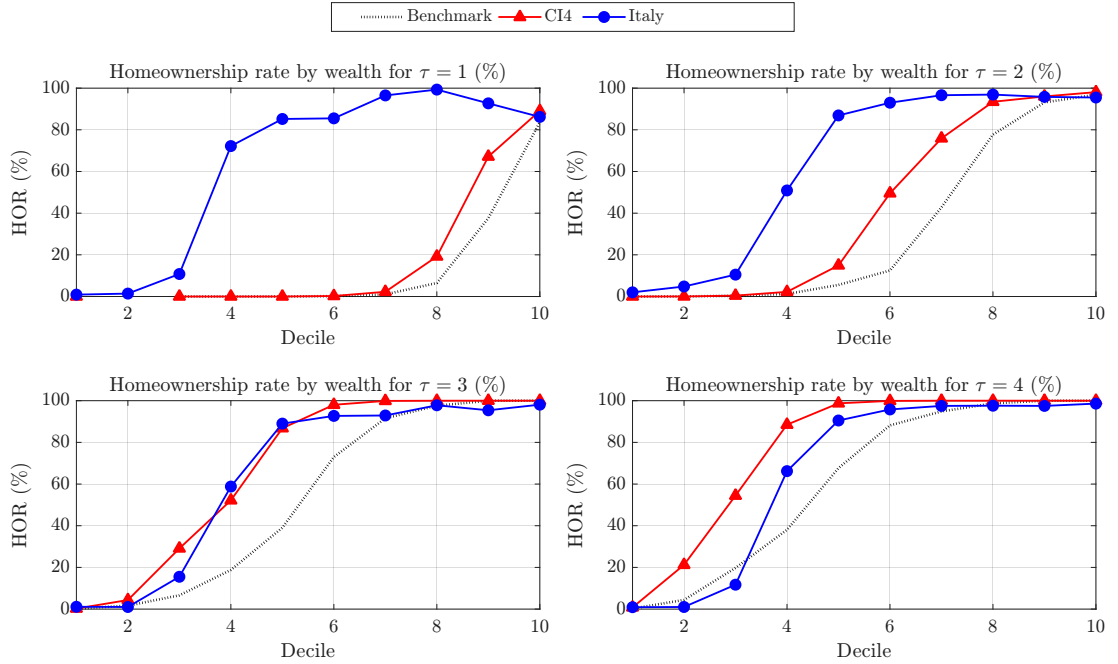


Figure A5: Homeownership rate by wealth and age group: Germany vs Italy vs CI4 (GE)

A.1 RTB Mechanism Diagnostics

This subsection documents all the diagnostic tables concerning the impact of the RTB mechanism discussed in Section 6.2: the RTB take up rates (Table A3), the social housing composition (Table A4), the reduction in housing consumption upon exercise (Table A5), and the wealth reclassification of RTB purchasers (Table A6). The last two tables refer only to CI2, whereas all tables (except for Table A5) are in GE. Similar patterns for these results can be found also in the not-presented specifications.

Table A3: RTB take-up rates (GE).

<i>Panel A: RTB take-up rate by age (%)</i>										
	25–34	35–44	45–54	55–64	65+	All				
CI2 (RTB only)	65.2	80.0	87.6	88.2	55.3	76.3				
CI3 (Italian SH)	52.1	68.0	79.5	79.1	38.9	63.9				
CI4 (Combined)	51.8	68.7	78.0	78.5	36.1	63.2				

<i>Panel B: RTB take-up rate by wealth decile D (%), all ages</i>										
	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
CI2 (RTB only)	22.5	52.1	94.2	100.0	100.0	100.0	100.0	100.0	100.0	100.0
CI3 (Italian SH)	6.4	42.0	88.2	100.0	100.0	100.0	100.0	100.0	100.0	—
CI4 (Combined)	6.6	47.1	90.2	100.0	100.0	100.0	100.0	100.0	100.0	—

Notes. Take-up rate is the fraction of social housing tenants who would exercise the RTB option if offered. Panel B uses wealth decile cutoffs from the overall benchmark population (all ages, all tenure types). A dash (—) indicates that no social housing tenants fall in that wealth decile, so no take-up rate can be computed.

Table A4: Social housing composition (GE).

<i>Panel A: SH tenant distribution by age (%)</i>							
	25–34	35–44	45–54	55–64	65+	(% pop)	
Benchmark	12.7	17.8	19.1	20.2	30.1	7.11	
CI2 (RTB only)	30.9	27.6	18.7	15.3	7.4	1.39	
CI3 (Italian SH)	31.8	28.1	16.9	14.4	8.8	1.06	
CI4 (Combined)	33.5	28.4	16.7	13.1	8.3	0.98	

<i>Panel B: SH tenant distribution by wealth decile D (%), all ages</i>										
	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
Benchmark	17.2	9.2	18.0	14.5	11.4	10.9	9.8	6.5	2.4	0.1
CI2 (RTB only)	12.5	14.3	31.1	16.8	11.1	8.3	4.1	1.4	0.4	0.0
CI3 (Italian SH)	18.4	19.9	29.8	14.5	8.5	5.2	2.6	0.9	0.2	0.0
CI4 (Combined)	19.3	21.2	30.8	14.2	7.7	3.8	2.0	0.8	0.2	0.0

Notes. Panel A shows the age distribution within the social housing tenant population. (% pop) is the fraction of the overall population living in social housing. Panel B uses wealth decile cutoffs from the overall benchmark population (all ages, all tenure types).

Table A5: Housing services: Social housing tenants vs. RTB purchasers (CI2, PE).

Age	Avg s (tenant)	Avg h (RTB)	Ratio h/s	RTB mass
25–34	159.1	104.5	0.657	0.0006
35–44	179.6	122.1	0.680	0.0006
45–54	173.1	130.5	0.754	0.0005
55–64	153.3	128.0	0.835	0.0004
65+	93.5	93.7	1.002	0.0001
All	161.6	118.8	0.735	0.0021

Notes. Avg s (tenant) is the mass-weighted average housing services consumed by social housing tenants ($\sigma = 1$). Avg h (RTB) is the mass-weighted average house size chosen by those who exercise the RTB option, where housing consumption is fixed at $s = h$ in the locked-in state. RTB mass is the per-period flow of exercising households. The ratio $h/s < 1$ for working-age groups confirms that RTB purchasers buy smaller units than they were renting, explaining the aggregate housing demand contraction documented in Table 6. The ratio is approximately unity for retirees, reflecting the fact that the elderly tenants who can afford to exercise tend to have modest rental consumption, so the purchased unit roughly matches their pre-exercise housing services.

Table A6: RTB wealth transition matrix (CI2, GE).

<i>Pre-exercise $\rightarrow$ post-exercise wealth decile D (% of row mass)</i>											
Pre \ Post	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Mass (%)
D1	.	.	.	.	99.8	0.2	.	.	.	.	0.15
D2	.	.	.	.	99.9	0.1	.	.	.	.	5.30
D3	.	.	.	.	91.8	7.9	.	0.4	.	.	36.81
D4	.	.	.	.	6.6	77.8	.	15.3	0.2	.	21.37
D5	.	.	.	.	.	25.0	15.7	50.6	8.6	.	14.15
D6	.	.	.	.	.	.	23.3	4.5	72.2	.	10.37
D7	.	.	.	.	.	.	3.6	4.7	90.5	1.2	6.25
D8	.	.	.	.	.	.	.	0.6	45.9	53.5	2.93
D9	.	.	.	.	.	.	.	.	.	100.0	1.92
D10	.	.	.	.	.	.	.	.	.	100.0	0.76

Notes. Each row shows how RTB exercisers originating from a given pre-exercise wealth decile are distributed across post-exercise wealth deciles. Pre-exercise net worth is financial assets only (a), since tenants have $h = 0$. Post-exercise net worth is $a' + p \cdot h_{\text{RTB}}$, where the house enters the balance sheet at market value. Mass (%) is the fraction of all RTB exercisers originating from each pre-exercise decile. Dots indicate entries below 0.05%. The average wealth jump upon exercise is 75.0 thousand euros (from 68.1 to 143.1), driven almost entirely by the market valuation of the acquired house. Wealth decile cutoffs are from the benchmark overall population.

B Real-Estate Firm Zero-Profit Condition under RTB

This appendix derives the zero-profit condition of the real-estate firm operating a social housing unit under the RTB mechanism, and shows that the government subsidy described in Section 5.2 preserves (7) in its original form.

Consider a real-estate firm holding a social housing unit, worth p^s on its balance sheet. In each period, the unit generates rental income ρ^s and imposes monitoring cost c^m on the firm. The unit depreciates at rate δ . In the baseline model, the unit exits the social housing sector only through reconversion to the regular market, which occurs with probability Φ and at which point the firm receives p in cash while losing the asset worth p^s .

Introducing the RTB mechanism adds a second exit channel. Let $\bar{\pi}^{RTB}$ denote the population-weighted probability that a social housing unit held by the firm is transferred to its tenant through an RTB exercise in a given period. When this event materializes, the household pays the firm $p(1 - d)$ and the government pays the firm $d \cdot p$ as a compensating subsidy, so that the firm's total receipt is p . The unit then leaves the firm's balance sheet, just as under reconversion.

The no-arbitrage condition for the firm therefore becomes

$$rp^s = \underbrace{\rho^s - c^m}_{\text{rent net of monitoring}} - \underbrace{\delta p^s}_{\text{depreciation}} + \underbrace{\Phi(p - p^s)}_{\text{capital gain from reconversion}} + \underbrace{\bar{\pi}^{RTB}(p - p^s)}_{\text{capital gain from RTB}} \quad (39)$$

The RTB term enters symmetrically to the reconversion term. Within the expected capital gain bracket $(p - p^s)$, the negative component captures the outflow of the asset (worth p^s) from the firm's balance sheet when the unit is purchased, while the positive component (p) captures the total cash inflow that compensates the firm for the loss. The difference $p - p^s$ is valued positively because social housing units are priced below market ($p^s < p$) to reflect the firm's commitment to rent them at the subsidized rate ρ^s .

Rearranging (39):

$$(r + \delta + \Phi + \bar{\pi}^{RTB}) p^s = \rho^s - c^m + (\Phi + \bar{\pi}^{RTB}) p \quad (40)$$

Solving for p^s :

$$p^s = \frac{\rho^s - c^m + (\Phi + \bar{\pi}^{RTB}) p}{r + \delta + \Phi + \bar{\pi}^{RTB}} \quad (41)$$

(41) differs from the baseline formula

$$p_{\text{baseline}}^s = \frac{\rho^s + \Phi p - c^m}{r + \Phi + \delta} \quad (42)$$

only through the additional $\bar{\pi}^{RTB}$ terms in the numerator and denominator.

A direct algebraic comparison of (41) and (42) shows that p^s is unchanged to the addition of the RTB channel if and only if

$$\frac{\rho^s + (\Phi + \bar{\pi}^{RTB})p - c^m}{r + \delta + \Phi + \bar{\pi}^{RTB}} = \frac{\rho^s + \Phi p - c^m}{r + \Phi + \delta} \quad (43)$$

Cross-multiplying and simplifying, this identity holds precisely when

$$\rho^s - c^m + \Phi p = (r + \delta + \Phi) p^s, \quad (44)$$

which is exactly the baseline zero-profit condition (7). In other words, if p^s is pinned down by (7) in the baseline, it remains pinned down by the same equation under RTB, because adding the same $\bar{\pi}^{RTB}$ term to the numerator and denominator of (41) leaves the ratio unchanged.

The mechanism behind this invariance is economically intuitive. By paying the firm $d \cdot p$ per RTB transaction, the government ensures that the firm receives the same cash amount (p) whether the unit exits via reconversion or via RTB. From the firm's perspective, the two exit channels are then perfectly substitutable in their cash-flow implications, so adding RTB as a second exit channel alters neither the expected cash flow per unit nor the equilibrium price p^s .

The logic of the RTB subsidy mirrors the construction subsidy ς introduced in Section 4. In the baseline, the construction firm's marginal cost equals p , while social housing firms are willing to pay only p^s for a unit given their restricted rental revenue. The government closes this gap by paying the construction firm $\varsigma = p - p^s$ per social housing unit built, so that the construction firm is indifferent between supplying the regular market and the social housing sector (9).

C RTB Mechanism, free unit allocation

While the institutional reality of Legge 560/1993 dictates that tenants purchase the exact physical unit they currently occupy, implementing this in the model at hand would imply introducing an additional state variable. Following Kaas et al. (2021), social housing tenants freely re-optimize their housing services s each period as a flow decision. There is no persistent dwelling assignment that tracks which apartment the household was allocated upon entry. In reality, social housing tenants are assigned a specific apartment and remain in it unless administratively relocated, so their housing consumption is largely fixed by the initial assignment rather than re-optimized each period. Introducing such a persistent assignment would require adding a dwelling-size state variable to the social housing tenant's problem, expanding the state space in a way that is beyond the scope of this thesis, beside the need to recalibrate the baseline model in order to account for this change. The free-choice formulation adopted here should therefore be understood as a reduced-form approximation that captures the economic core of the RTB mechanism while abstracting from the institutional rigidity of apartment assignments. This formulation also grants the tenant more flexibility than the real institution provides: in particular, a tenant currently consuming large housing services at subsidized rent can opt to purchase a smaller, more affordable unit. This flexibility likely overstates RTB take-up at the margin, because some tenants who could not afford their current unit can instead buy a smaller one. At the same time, it understates the housing consumption of RTB owners relative to the institutional counterfactual, since many buyers in the model choose units smaller than what they were renting (see Table A6).

D Resale Ban Considerations

The RTB mechanism is modeled with a permanent lock-in: a household that exercises the option cannot subsequently resell, downsize, or rent out the unit. A natural question is how the results would change under a less restrictive assumption. Suppose that RTB owners faced a per-period probability $\pi_{unlock} = 0.1$ of regaining the ability to sell, matching the 10-year statutory restriction of Legge 560/1993. Two effects would operate in opposite directions. The value of exercising RTB would rise, because the locked-in state would be temporary rather than permanent, and with it the non-negativity constraint on financial assets would eventually be relaxed. This would make the option more attractive and potentially raise take-up. On the other hand, some freed RTB owners would choose to sell and revert to renting, particularly those with low financial assets for whom the costs of ownership outweigh the benefits. The net effect on homeownership would depend on the relative strength of these two channels.

Even under this relaxation, the model would not capture the complexity of the Italian resale regime, which includes caps on resale prices, rights of first refusal for the public authority, and extinction clauses tied to cadastral-value reimbursements after the restriction period. A faithful representation of these features would at the same time introduce further state variables and transition rules. Beyond institutional detail, a finite lock-in specification also raises a computational hindrance which lies beyond the scope of this thesis. Under probabilistic transitioning to resale freedom, the value of the locked-in state V^{lock} would depend on the expected value of becoming a standard owner in the private market, V^0 , through the prospect of unlocking. However, V^0 itself depends on the option to enter social housing, V^1 , since eligible renters may choose to do so; the value of social housing tenancy depends on the prospect of exercising the RTB option, V^{RTB} ; and V^{RTB} leads back to V^{lock} . This circularity can be resolved through an iterative procedure such as a two-pass Gauss-Seidel scheme (Walker, Auerbach, and Kotlikoff 1988), but the additional layer of complexity would come at a computational cost which might not necessarily reflect the marginal gains from identifying the pure effect of the RTB mechanism on homeownership, which is the principal object of interest in this experiment.

E Counterfactual with fixed social housing stock

As mentioned in Section 5.2, the implementation of the RTB mechanism inside this thesis project has come with the decision to let the social housing stock adjust endogenously, with the government deliberately not replacing the lost housing units with new ones. This choice, while keeping the policy analysis consistent with the idea to change only specific parameters while letting other objects adjust, has also the downside of generating a conservative estimate of the RTB impact, compared to how it evolved in reality in Italy. In this appendix I thus explore the opposite extreme, in which the social housing stock is held at its benchmark level of 7.1% across all four counterfactuals. This is performed by adjusting the entry probability π (and implicitly the social housing construction subsidy ς).

For CI1 the implications of this design are minor: in the absence of the RTB mechanism, the only reason the stock moves in the endogenous case is a mild private-market pull induced by the lower RETT, so holding the stock fixed requires only a modest adjustment of π . For the three counterfactuals that include RTB (CI2, CI3, CI4), the implications are substantial: every tenure conversion through RTB is a permanent outflow from the social housing sector, and replacing these units requires a considerably larger per-period inflow.

Table E1 reports the access probability and the resulting stock share under the two specifications. For CI1, π rises by a factor of 1.26, in line with the corresponding adjustments in the Kaas et al. (2021)'s replication of their C1 experiment. For CI2, CI3 and CI4, π must rise by factors between 7.9 and 11.5. The CI3 multiplier (10.8) is larger than the CI2 one (7.9) because the restricted eligibility concentrates the inflow on the bottom three income deciles: to replace the same number of RTB exits from a smaller eligible pool, the access probability within that pool must be correspondingly higher. The CI4 multiplier (11.5) is the largest because the RETT cut additionally pulls a marginal fraction of tenants into the private market. Having such high entry rates also implies that the turnover for eligible households to be actually selected is quite high, increasing consequently also the chance that more households get access to the RTB option, thus increasing homeownership rates.

Table E1: Access probability and social housing stock: endogenous vs. fixed stock (PE).

<i>Panel A: Access probability (%)</i>				
	CI1	CI2	CI3	CI4
Endogenous stock	1.28	1.28	1.28	1.28
Fixed stock (7.1%)	1.61	10.11	13.84	14.68
Multiplier (fixed/endog.)	1.26×	7.90×	10.81×	11.47×

<i>Panel B: Social housing stock share (%)</i>				
	CI1	CI2	CI3	CI4
Endogenous stock	5.77	1.42	1.06	0.98
Fixed stock	7.1	7.1	7.1	7.1

Notes. Benchmark values: access probability $\pi = 1.28\%$, social housing stock share = 7.11%. The access probability is $\pi_T(y)$ evaluated at an eligible state; in CI3 and CI4 only the bottom three income deciles are eligible, so $\pi_T(y) = 0$ for the top seven deciles. The multiplier reports the factor by which π is rescaled to hold the social housing stock share at the benchmark 7.1% given the RTB-induced outflow. All results are under partial equilibrium (fixed prices, fixed taxes).

Figures E1 and E2 compare the two specifications. Under CI1, the two curves track each other closely across both the age and wealth dimensions, with the fixed-stock case lying uniformly 0.5 to 2.5 percentage points below the endogenous one: by preserving a larger social housing sector, fewer households are pushed into the private market, which marginally reduces the homeownership response. Under CI2, CI3 and CI4 the pattern changes substantially. Along the life cycle, the fixed-stock specification delivers homeownership gains everywhere, though most evident in two regions, common to all three experiments: at working ages 35 to 44, where cohorts now face a much higher probability of entering social housing and accumulating RTB exposure along their life-cycle, homeownership gains lie between 4 and 11 percentage points; and at age 65 and above, where the cumulative stock of past RTB exercises reaches 15 to 18

percentage points above the endogenous case. At middle working ages (45 to 54 and 55 to 64) the pattern is more nuanced: while positive gains are registered in CI2 and CI3, in CI4, the fixed-stock homeownership rate falls slightly below the endogenous one.

Along the wealth distribution, the differences are more pronounced and shaped by the interaction of two mechanisms. Wealth of locked-in households is structurally bounded, with financial assets constrained to be non-negative and housing capped at the social housing size limit. This means that the mass of locked-in owners concentrates in the middle of the working-age wealth distribution. Because wealth deciles are computed on each economy's own working-age distribution, the large locked-in cluster also redefines the cutoffs: the lower boundaries of the middle deciles shift upwards to accommodate the locked-in band, while the top-decile thresholds compress downwards. This redefinition accounts for most of the apparent decline in homeownership at deciles 7 to 10 in the fixed-stock specification, as now households that would have been placed at lower deciles of the distribution now belong to the upper tail. Finally, another effect operates alongside: with a much higher access probability, eligible households cycle through the social housing sector as renters while accumulating substantial financial wealth from the subsidized rent. A fraction of them thus reach the upper tail of the distribution as renters, either because the right-to-buy offer does not arrive in time or because their desired housing exceeds the social housing size cap.

Table E2: Working-age wealth decile cutoffs (thousand euros, PE)

Specification	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
Benchmark (Germany)	16.4	28.5	52.7	70.9	101.2	134.0	181.7	240.8	327.6	980.0
CI1										
Endogenous stock	16.4	34.5	52.7	77.0	112.3	155.1	204.4	263.7	353.8	980.0
Fixed stock (7.1%)	16.4	34.5	52.7	77.0	110.9	154.4	203.6	263.6	353.7	980.0
Difference	+0.0	+0.0	+0.0	+0.0	-1.4	-0.8	-0.8	-0.1	-0.1	+0.0
CI2										
Endogenous stock	16.4	34.5	52.7	83.0	107.3	137.6	177.6	238.2	323.1	980.0
Fixed stock (7.1%)	22.4	40.6	83.0	101.2	120.6	150.9	186.1	234.5	303.3	980.0
Difference	+6.1	+6.1	+30.3	+18.2	+13.3	+13.3	+8.5	-3.7	-19.7	+0.0
CI3										
Endogenous stock	16.4	34.5	52.7	77.0	104.9	137.0	180.0	240.0	324.4	980.0
Fixed stock (7.1%)	16.4	40.6	70.9	96.4	119.4	144.8	180.0	228.5	302.4	980.0
Difference	+0.0	+6.1	+18.2	+19.4	+14.5	+7.9	+0.0	-11.5	-22.0	+0.0
CI4										
Endogenous stock	16.4	34.5	52.7	83.0	109.1	149.7	199.9	258.8	345.8	980.0
Fixed stock (7.1%)	16.4	40.6	77.0	96.4	120.6	149.8	190.9	244.3	319.5	980.0
Difference	+0.0	+6.1	+24.2	+13.3	+11.5	+0.2	-9.1	-14.5	-26.2	+0.0

Notes: Values are reported in thousands of 2006 euros. Difference is calculated as Fixed stock minus Endogenous stock. All specifications assume partial equilibrium (PE).

Taken together, the endogenous and fixed-stock specifications bracket the plausible range of the RTB mechanism's contribution to homeownership. The endogenous case likely understates the true effect, because it lets the pool of eligible tenants shrink without replacement and thereby attenuates the RTB outflow over time. The fixed-stock case likely overstates it, because it would

require the government to increase access to social housing by roughly an order of magnitude, an intervention that would itself have constituted a major policy intervention rather than a neutral backdrop for the RTB mechanism. The magnitude of the homeownership response under the true historical trajectory should therefore lie between the conservative estimate in the main text and the upper bound reported here.

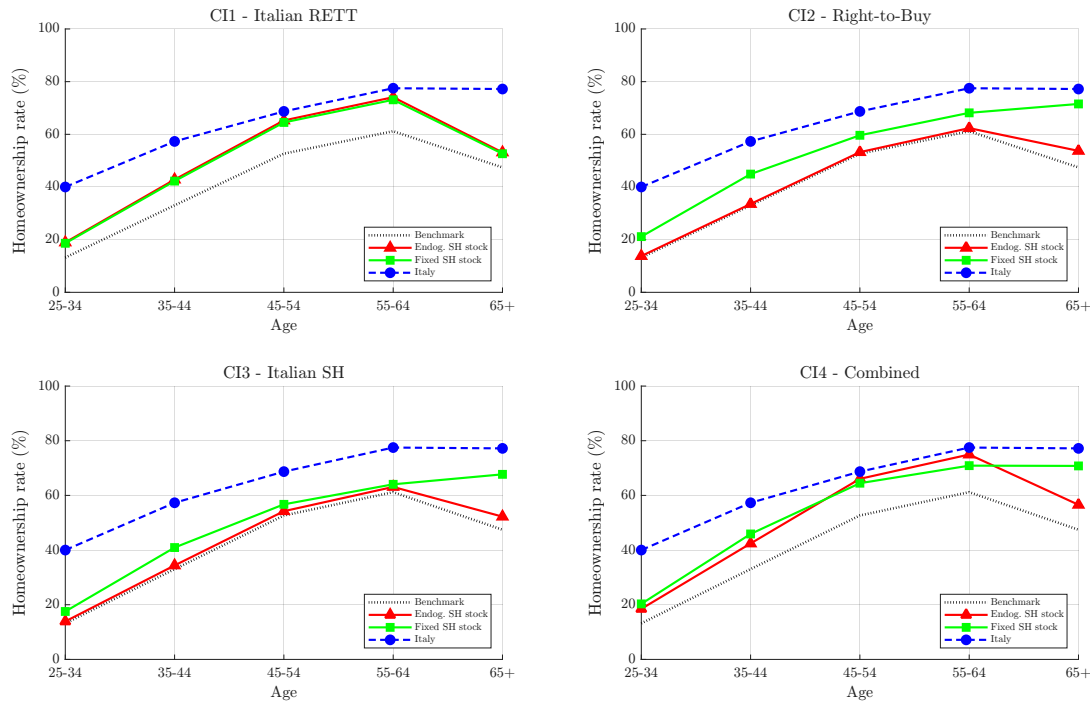


Figure E1: Homeownership rate by age group: Germany vs Italy vs CI4 (PE)

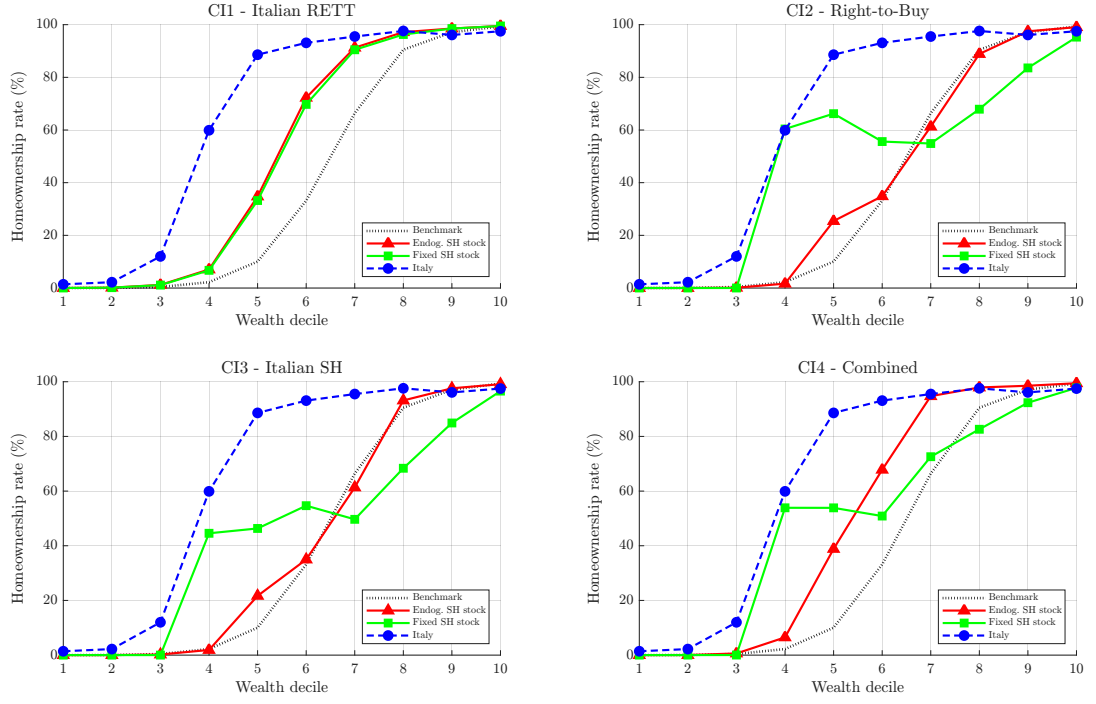


Figure E2: Homeownership rate by wealth decile (working age group): Germany vs Italy vs CI4 (PE)

F Sensitivity Analysis of the Italian Social Housing Parameters

The three parameters governing the structure of the Italian social housing system in the counterfactuals of Section 5, i.e. the RTB discount $d = 0.68$, the eligibility threshold at the bottom three income deciles, and the 40% rent discount, are grounded in the institutional and empirical evidence reviewed in Section 3. Each of them is nonetheless subject to some measurement uncertainty, given the absence of a unique estimate in the literature. This appendix thus reports a sensitivity analysis that perturbs one of these parameters at a time around its baseline value, keeping all other elements of the environment fixed, so as to verify how much do the baseline responses change depending on the magnitude of these coefficients. The RTB discount is varied within the CI2 environment, while the eligibility threshold and the rent discount are varied within the CI3 environment. All experiments are conducted in partial equilibrium, that is, holding the house price and the income tax scale at their benchmark values, so that the comparative statics isolate the direct effect of each parameter on household behavior.

F.1 RTB Discount

Table F1 reports the CI2 results for $d = 0.68, 0.75$, and 0.84 , spanning the broad interval stated by Bianchi (2014). The aggregate homeownership rate is flat across the three specifications, moving only between 45.06 and 45.13 percent. What responds to the discount are the intensive margins within the social housing sector: the take-up rate rises from 76.2 to 90.9 percent, the locked-in share from 9.07 to 10.32 percent, and the social housing stock falls from 1.42 to 1.17

percent of the population.

The near invariance of aggregate homeownership masks an offset between two channels. Decomposing ownership into locked-in RTB buyers and regular private-market owners, the 1.25 pp increase in the locked-in share between $d = 0.68$ and $d = 0.84$ is matched by a 1.23 pp decline in regular ownership (from 35.99 to 34.76). The deeper discount, rather than attracting many more new homeowners into the economy, redirects the tenure outcome of households who, under the baseline, would have accumulated sufficient wealth to exit social housing and purchase a unit on the private market. For these households, the RTB option becomes preferable to the exit-and-purchase alternative, and they remain in the sector long enough to exercise the discounted purchase. The channel through which households become owners shifts from the private market to the RTB path, but the gross flow from tenancy to ownership is nearly unchanged.

The implication is that the size of the RTB discount governs the composition of homeownership among former social housing tenants rather than the extensive margin of ownership. At $d = 0.68$ the take-up rate is already 76 percent, indicating that the self-financing constraint is slack for the majority of tenants who find the option attractive. Further deepening of the discount operates primarily as a substitution between tenure channels, with a crowding-out effect on the private-market route that approximately cancels the expansion of the RTB option.

Table F1: Sensitivity of CI2 to the RTB discount d (PE).

Specification	HOR	25–34	35–44	45–54	55–64	65+	SH%	L.I.%	T-up%	Gap%
Benchmark (Germany)	42.49	13.2	33.0	52.7	61.2	47.4	7.11	0.00	–	–
CI2 baseline ($d = 0.68$)	45.06	13.8	33.5	53.2	62.4	53.7	1.42	9.07	76.2	9.8
CI2 $d = 0.75$	45.13	13.9	33.6	53.1	62.0	54.1	1.31	9.57	82.2	10.1
CI2 $d = 0.84$	45.08	14.0	33.3	52.5	61.4	54.6	1.17	10.32	90.9	9.9

Notes. All specifications are in partial equilibrium (p and λ_T at benchmark). HOR denotes the aggregate homeownership rate. Age columns report HOR by age bin. SH% is the social housing stock share. L.I. is the share of households in the locked-in state. T-up% is the aggregate RTB take-up rate, weighted by the mass of eligible households across ages. Gap% is the fraction of the Germany–Italy HOR gap closed, with $\text{HOR}_{IT} = 68.7$ percent.

F.2 Eligibility Threshold

One could also argue that, given the residualistic nature of the Italian social housing system, giving access to the third income decile might not be truly realistic. Table F2 thus narrows the eligibility set from the bottom three to the bottom two income deciles, holding all other CI3 parameters fixed. Aggregate homeownership rises from 45.01 to 45.44 percent, closing 11.3 percent of the gap against Italy rather than 9.6. All social housing indicators contract simultaneously: the stock share falls from 1.06 to 0.82 percent, the locked-in share from 5.47 to 3.68 percent, and the take-up rate from 64.1 to 57.1 percent.

The mechanism is the displacement of households in the excluded third decile, who under the broader rule would have had positive probability of entering the social housing sector. With tighter targeting, these households face the private rental market directly from the beginning of the working life, and a fraction of them accumulate sufficient wealth to transition into regular ownership. The simultaneous decline in take-up is informative about who remains eligible under the narrower rule. The surviving pool is on average poorer, so a smaller fraction can clear the

down-payment required to exercise the RTB option, even at the 68 percent discount.

This implies that tighter targeting does not simply rescale the social housing sector uniformly, rather it shifts the balance between the two roles that the Italian system plays in the model: the tenancy function becomes more prominent while the pathway to ownership function recedes, even though the formal RTB rules are identical across the two specifications.

Table F2: Sensitivity of CI3 to the eligibility threshold (PE).

Specification	HOR	25–34	35–44	45–54	55–64	65+	SH%	L.I.%	T-up%	Gap%
Benchmark (Germany)	42.49	13.2	33.0	52.7	61.2	47.4	7.11	0.00	–	–
CI3 baseline (bottom 3 deciles)	45.01	13.9	34.4	54.2	63.1	52.2	1.06	5.47	64.1	9.6
CI3 bottom 2 deciles	45.44	14.2	35.2	55.7	64.3	51.6	0.82	3.68	57.1	11.3

Notes. All specifications are in partial equilibrium. Column definitions as in Table F1. The eligibility threshold is the set of income deciles that receive positive entry probability into the social housing sector, as described in Section 5.3

F.3 Rent Discount

Table F3 varies the rent discount along the CI3 path, holding eligibility fixed at the bottom three deciles, spanning the discount values presented by Baldini and Poggio (2012). The baseline specification has a social housing rent with a 40 percent rent discount relative to the private market. Deepening the discount to 60 percent reduces homeownership rate from 45.01 to 44.41, and deepening it further to 80 percent reduces it to 42.79, close to the benchmark German level. The social housing stock expands from 1.06 to 2.46 percent over the same range, while the take-up rate collapses from 64.1 to 23.6 percent.

The mechanism is a substitution between the two subsidy instruments embedded in the Italian system. The rent discount acts as a flow subsidy conditional on remaining a tenant, while the RTB acts as a stock subsidy conditional on exiting into ownership. When the rent discount is moderate, tenants find the exit option relatively more valuable and exercise the RTB when it arrives. When the rent discount is deep, the option value of remaining a tenant rises, and the continuation value of social housing dominates the discounted purchase value. Households stay in the sector rather than exercising, take-up falls, and the locked-in share declines despite the larger pool of eligible households.

The response at 80% discount is exemplary. Under this specification, the RTB channel is almost entirely neutralized, with a take-up of 23.6 percent and a gap closed of only 1.2 percent, even though the RTB rules and the eligibility threshold are identical to the baseline. The rent discount parameter therefore dominates the RTB discount parameter in shaping the aggregate tenure outcome under the Italian design. The effectiveness of the RTB mechanism documented in CI3 and CI4 is thus conditional on the magnitude of the rent subsidy.

Table F3: Sensitivity of CI3 to the rent discount ρ^s (PE).

Specification	HOR	25–34	35–44	45–54	55–64	65+	SH%	L.I.%	T-up%	Gap%
Benchmark (Germany)	42.49	13.2	33.0	52.7	61.2	47.4	7.11	0.00	–	–
CI3 baseline with 40% discount	45.01	13.9	34.4	54.2	63.1	52.2	1.06	5.47	64.1	9.6
CI3 with 60% discount	44.41	13.3	33.4	53.3	62.5	52.0	1.22	5.42	55.8	7.3
CI3 with 80% discount	42.79	11.7	30.7	50.5	60.9	51.5	2.46	4.25	23.6	1.2

Notes. All specifications are in partial equilibrium. Column definitions as in Table F1. The discount percentage in the row label refers to the reduction of the social housing rent relative to the private market rent ρ , with $\rho^s = (1 - \text{discount})\rho$.

G Data and Methodology

This appendix outlines the data sources, sample selection, and empirical methodology used to calculate the homeownership rates presented in the thesis. Two datasets are involved: the German Socio-Economic Panel (SOEP) for the benchmark economy, and the Italian Survey on Household Income and Wealth (SHIW, Banca d’Italia (2025)) for the cross-country comparison.

G.1 German Data

The empirical facts on German homeownership, income, and wealth used by Kaas et al. (2021) are derived from the SOEP. The sample pools the wealth modules collected in 2002, 2007, and 2012, restricted to households whose head is aged 25 or older. Business owners are excluded to maintain consistency with the model, which does not feature entrepreneurship. Working-age household heads are further required to have positive labor income. Homeownership is defined as owning the primary household residence. Household net wealth is defined as the value of all real and financial assets net of liabilities. All monetary values are CPI-deflated and expressed in 2006 euros. The resulting pooled dataset consists of 24,595 households, yielding an aggregate homeownership rate of approximately 44% for households with heads aged 25 or older. These data moments are used both for calibration and for the model validation exercise.

G.2 Italian Data

The Italian empirical analysis relies on microdata from the Survey on Household Income and Wealth (SHIW), conducted by the Bank of Italy and structured according to the standardized Household Finance and Consumption Survey (HFCS) format. I pool cross-sectional data from two survey waves: 2010 and 2014. For each wave, I extract variables from the first implicate³⁴ (IM0100 = 1) across three primary files:

- **Personal Core (p1):** Individual-level demographics, including the relationship to the reference person (ra0100) and age (ra0300).
- **Household Core (h1):** Household-level variables, notably the tenure status of the main residence (hb0300) and the household survey weight (hw0010).

³⁴This represents the initial complete dataset generated by replacing missing values with a set of plausible values by the Bank of Italy.

- **Derived Variables (d1):** Aggregated financial metrics, including total household gross income (di2000) and household net wealth (dn3001), defined as total assets excluding public and occupational pension wealth minus total liabilities.

G.3 Sample Selection and Variable Construction

The sample is restricted to the household Reference Person (identified by ra0100 == 1) and observations where the reference person is under 25 years of age are dropped. Age is categorized into five bins: 25–34, 35–44, 45–54, 55–64, and 65 or older. The primary variable of interest, homeownership, is defined as a binary indicator equal to 1 if the household legally owns the entirety or a part of their main residence (hb0300 ∈ {1, 2}), and 0 for all other arrangements.

For the distributional analysis by wealth, net wealth (dn3001) is first deflated to 2006 euros using the Italian Harmonized Index of Consumer Prices (HICP, (Eurostat 2025)), consistent with the deflation applied by Kaas et al. (2021) to the German SOEP data. Table G1 reports the wave-specific deflators. The deflated values are then converted from euros to thousands of euros for comparability with the model’s units. Wealth deciles are computed within the Italian working-age (25–64) population using survey weights, so that each decile contains exactly 10% of working-age Italian households. This approach, which mirrors how Kaas et al. (2021) compute wealth deciles from the model’s stationary distribution, ensures that each decile label represents the same relative position within the respective country’s wealth distribution rather than the same absolute wealth level. The homeownership rate by age is computed for all households aged 25 and above (including retirees), while the homeownership rate by wealth decile is computed for working-age households only (25–64).

Table G1: HICP deflators for Italian net wealth

Wave	HICP (2015=100)	Deflator (HICP/85.6)
2006 (base)	85.6	1.000
2010	92.6	1.082
2014	99.9	1.167

Notes: HICP values are annual averages from Eurostat (prc_hicp_aind), Italy, all items, index 2015=100. Each wave’s net wealth (dn3001) is divided by the corresponding deflator to express values in 2006 euros.

G.4 Pooling and Weighting

Following the decile assignments, the two cross-sectional waves are appended into a single pooled dataset. In the SHIW framework, the sum of household weights in any single wave equals the total target population of Italian households for that year. Stacking two waves therefore results in an artificially inflated aggregate population. To normalize the representative population back to a single average cross-section spanning the 2010–2014 period, I construct a pooled weight:

$$w_{\text{pooled}} = \frac{hw0010}{2} \quad (45)$$

The average homeownership rate for the pooled sample under these specifications is 68.7%, closely aligned to the estimates of Grevenbrock, Ludwig, and Siassi (2024) for 2010 data.

G.5 Differences from the German Sample

Two methodological differences between the German and Italian samples are worth noting. First, Kaas et al. (2021) exclude business owners from the SOEP to maintain consistency with a model that does not feature entrepreneurship. The Italian sample retains all households, reflecting the structurally higher prevalence of self-employment in Italy (approximately 23% of employment, compared to roughly 10% in Germany). Excluding Italian self-employed households would remove a large and distinctive segment of the economy and would make the resulting homeownership rates unrepresentative of the country the thesis compares against. Second, while both samples define the unit of observation as the household and assign age based on the household head (SOEP) or Reference Person (SHIW), the SOEP household head is always defined as the male partner in married couples, whereas the SHIW Reference Person is the financially knowledgeable person regardless of gender. This difference is unlikely to affect the results. In both samples, homeownership is defined as owning the primary residence, and net wealth is defined as total assets (excluding public pension wealth) minus total liabilities.

H Computational Procedure

The following appendix describes the computational procedure used to solve the model described in this thesis. It differentiates between the baseline model, whose resolution relies on the work of Kaas et al. (2021), and the extensions made by the author of this thesis to produce the counterfactual experiments seen in the main text.

H.1 Baseline Computational Procedure

The household problem involves a joint discrete-continuous optimization: each period, households choose a discrete housing tenure (rent, own, or social housing) alongside continuous decisions over financial assets, housing size, and housing services consumption. Using the work of Kaas et al. (2021), the model is solved on a discretised state space using value function iteration (VFI), accelerated by Howard policy iteration (Howard 1960; Puterman 1994).

Discretisation. The state space $(a, i, h, \rho, \sigma, \tau)$ is first discretised with $n_a = 100$ uniformly spaced asset grid points on $[-220, 380]$ (thousand euros), $n_{ho} = 33$ housing grid points (of which $n_{ro} = 3$ represent renter states with idiosyncratic rental risk and 30 represent owned housing on a quadratic grid from 80 to 600), and $n_e = 10$ income states per age group. The quadratic spacing of the housing grid places more points at smaller house sizes, where the curvature of the value function is higher, improving the accuracy of the discrete approximation in the region most relevant for first-time buyers.

The three stochastic processes in the model are discretised using different methods. The income process is estimated non-parametrically from the German Socio-Economic Panel (SOEP) using the procedure described in Kaas et al. (2021, Appendix A) following De Nardi, Fella, and Paz-Pardo (2020). For each age group, residual log earnings are sorted into ten equally sized bins, and the transition matrix elements are set to the observed share of households moving between bins across consecutive years. The resulting matrices are normalized to be doubly stochastic using the Sinkhorn-Knopp algorithm (Sinkhorn 1966; Knight 2008), ensuring that the stationary distribution is uniform across deciles, consistent with the decile-based construction.

Idiosyncratic house value shocks are discretised using a direct normal CDF integration method adapted for the non-uniform housing grid, in the spirit of Tauchen (1986).

Idiosyncratic rental risk is instead discretised using Tauchen (1986)’s method explicitly, applied to the AR(1) process for log rental rates. The resulting grid values are adjusted to ensure that the unconditional mean rental rate equals $\bar{p}$ exactly. For current renters, the rental rate transitions follow the discretised AR(1); for homeowners who transition to renting, the initial rental rate is drawn from the stationary distribution of the process. Taxes follow the log-linear specification of Heathcote, Storesletten, and Violante (2017).

Solution. The Bellman equation is solved backward over age ($\tau = 5, \dots, 1$), following the contraction mapping approach of Stokey, Lucas, and Prescott (1989). At each VFI iteration, expected continuation values incorporate income transitions, stochastic aging, social housing entry/exit, housing value shocks, and bequests for $\tau = 1, 2$. After each maximization step, 50 iterations of Howard’s policy improvement algorithm are applied: holding the policy $(a^*, \tilde{h}^*)$ fixed, the Bellman operator reduces to the linear map $V^{new} = u(a^*, \tilde{h}^*) + \beta \mathbb{E}[V^{old}]$, which can be evaluated by matrix multiplication without re-maximizing. This substantially accelerates convergence (Puterman 1994). The VFI terminates when two conditions are jointly satisfied: the discrete policy must be identical across consecutive iterations (no grid point switches for a' at any state), and the relative change in the value function must satisfy $\max |V^{new} - V^{old}| / (1 + |V^{old}|) < 10^{-6}$.

After the coarse-grid VFI converges, a fine-grid refinement step re-optimizes the continuous savings decision. For each state, a local grid of 20 points is constructed around the coarse solution and the continuation value is evaluated at these points via piecewise linear interpolation of the converged value function. Housing services are simultaneously re-optimized on a finer grid. The fine-grid solution is used for all subsequent computations.

Stationary distribution. The distribution of households $\mu(\rho, a, i, h, \sigma, \tau)$ is computed by non-stochastic forward iteration on the converged policy functions, following the approach standard in the heterogeneous-agent literature. It distinguishes two types of households: households in the private market (with mass f_τ^{s0}); households in the social housing system (with mass f_τ^{s1}). The distribution is iterated forward by applying in sequence: demographic transitions (stochastic aging, newborn entry, death of retirees), social housing flows (entry, exogenous and endogenous exit), endogenous household decisions over housing value shocks and bequest redistribution for $\tau = 1, 2$.

The forward iteration uses an inner-outer loop structure. Each outer iteration runs 50 inner steps before checking convergence, assessed as the maximum relative change in the distribution across all states, with tolerance 10^{-3} during the first pass and 10^{-5} during the second. An outer loop updates the endogenous bequest distribution: after the first pass through all ages, the wealth distribution of dying $\tau = 5$ households is recomputed, and $\tau = 1$ and $\tau = 2$ are re-solved before a second distribution computation.

Benchmark Equilibrium. A stationary equilibrium requires, among others, two conditions to hold: housing market clearing, meaning that the house price is consistent with the construction technology ($p = k_0(I + I^s)^\phi$); and government budget balance, meaning that tax revenues cover pensions, social housing subsidies, and public good expenditure ($G = T_{rev}^{inc} + T_{rev}^{RE} - Cost_{Pensions} - Cost_{SH}$). In the benchmark, the house price is normalized to $p = 1$ and the income tax function parameters are estimated directly from SOEP data for each age group. The model is then solved once (VFI, stationary distribution, aggregation). Two equilibrium objects are thus computed as residuals from the conditions above. The construction sector scale parameter k_0 is backed out from the housing market clearing condition given price normalization. Government consumption G_0 is instead obtained as the residual of the budget constraint. Both G_0 and k_0 are saved as part of the benchmark solution and held fixed in all subsequent counterfactual experiments.

H.2 Extensions for the Italian Counterfactuals

All counterfactual experiments depart from the solved German benchmark and modify policy parameters before re-solving the model. CI1 (Italian RETT) requires no algorithmic changes: the reduced transfer tax propagates through the budget constraint into the per-period return matrices. CI3 and CI4 additionally modify the social housing eligibility matrix (restricting access to the bottom three income deciles) and the subsidized rent level ($\rho^s = 0.6\bar{\rho}$), both of which enter the existing algorithm through the continuation values and return matrices, respectively.

The RTB mechanism (active in CI2, CI3, and CI4) is the sole computational extension. It introduces a new binary state variable $\mathbb{L} \in \{0, 1\}$ (lock-in status) and is implemented through a layered procedure that exploits the absorbing nature of the lock-in state. Because V^{lock} is self-contained, its continuation depends only on V^{lock} itself, so it can be solved before any other value function. Once V^{lock} is in hand, V^{RTB} can be computed using V^{lock} as continuation, again before the main VFI. Both objects are then injected into the standard VFI, where the social housing continuation is augmented to incorporate the option value of receiving an RTB offer next period. This layered design avoids any joint fixed point between V^{lock} , V^{RTB} and the standard value functions, and keeps the per-iteration cost of the main VFI close to the baseline.

Stage 1: Lock-in value function. The value function V^{lock} for locked-in RTB owners is solved independently by backward induction, before the main VFI. The lock-in constraints reduce the problem to a one-dimensional savings choice for each (i, h) . The same VFI algorithm as the baseline is used (Howard acceleration with 50 iterations, 10^{-6} convergence tolerance, fine-grid refinement with piecewise linear interpolation). The continuation values for V^{lock} account for income transitions, stochastic aging, housing value shocks and bequests for $\tau = 1, 2$. Social

housing transitions do not apply, since locked-in owners are outside the social housing system.

Stage 2: RTB value function. Once V^{lock} has converged, the value of exercising the RTB option, V^{RTB} , is computed for each social housing tenant state by enumeration over the feasible house sizes $\tilde{h}'$ and savings a' . The continuation $\mathbb{E}[V^{lock}]$ is constructed once per income state by integrating V^{lock} over the income transition matrix, the ageing probability ϑ_τ , the housing value shock, and (for $\tau = 1, 2$) the bequest distribution.

Within the RTB period itself, the optimization over housing services s admits a closed-form solution. The household has not yet occupied the purchased unit, which only arrives next period after the housing value shock, and continues to consume housing services from the social housing rental at rate ρ^s . The current-period budget constraint is $c + \rho^s s = W$, where $W = y(\tau, i) - T_\tau(y^i) + [1 + r\mathbb{I}_{a>0} + r^m\mathbb{I}_{a<0}]a - a' - p\tilde{h}'(1-d)(1+\tau^b)$ collects all resources net of the discounted purchase expenditure and the savings choice. This constraint is linear in s , with no rental income term and no kink at $s = h$. The Cobb-Douglas structure of the period utility then implies that the optimal expenditure share on services is a constant fraction of W (Mas-Colell, Whinston, and Green 1995, Ch. 3), giving $s^* = (1 - \zeta)W/\rho^s$ capped at the social housing size limit $\bar{s}$, and $c^* = W - \rho^s s^*$. The CRRA transformation does not affect the location of this optimum, since it is a monotone transformation of the objective. This closed-form solution replaces the inner grid search over services that the baseline applies for homeowners, and avoids an additional loop nested within the $(\tilde{h}', a')$ search.

Stage 3: Main VFI with RTB anticipation. The main VFI is run with social housing tenants anticipating the possibility of receiving an RTB offer next period. In each iteration of the main VFI, the continuation associated with retaining social housing access next period, which in the baseline takes the form $\Omega' = \max(V^{1'}, V^{0'})$, is replaced by

$$\Omega^{RTB'} = \max(V^{1'}, V^{0'}) + \pi_{RTB} \cdot \max(V^{RTB'} - \max(V^{1'}, V^{0'}), 0),$$

The corresponding bequest-augmented version, integrating $V^{RTB'}$ over the bequest distribution, is constructed analogously for $\tau = 1, 2$ and appears in the social housing branch of the continuation whenever the inheritance event materialises. Howard acceleration, fine-grid refinement and the second-pass re-solve of $\tau = 1, 2$ (after the bequest distribution has been updated by the first stationary distribution call) are kept identical to the baseline.

Stage 4: Take-up evaluation and augmented distribution. After the main VFI converges, the RTB take-up indicator is evaluated state by state by comparing the precomputed V^{RTB} with the converged anticipated value V^1 :

$$\mathbb{I}_{RTB}(\tau, i, \rho, \sigma, a, h) = \mathbb{I}[V^{RTB}(\tau, i, a) > V^1(\tau, i, \rho, \sigma, a, h)].$$

The stationary distribution is augmented with a third layer f_τ^{L1} for locked-in owners, initialised at zero. Three transitions involving the $L1$ layer are added to the baseline ones. First, inflows from social housing: at every state where $\mathbb{I}_{RTB} = 1$, a fraction π_{RTB} of the social housing mass flows into f_τ^{L1} at the RTB-policy positions $(a'_{RTB}, \tilde{h}'_{RTB})$, with off-grid allocation handled through the

same linear interpolation weights used elsewhere; the remaining $(1 - \pi_{RTB})$ fraction follows the standard social housing transition rules. Second, continuation of $L1$ mass: locked-in owners at transition according to the V^{lock} savings policy, with $h' = h$ enforced. Third, housing shocks for $L1$: the standard owner shock is applied to the $L1$ mass exactly as for private market households, and any mass whose housing unit shrinks below the renter threshold is expelled from $L1$ and added back to f_τ^{s0} , since the lock-in attaches to a unit that no longer exists. Bequests for $L1$ owners in $\tau = 1, 2$ are processed through the same bequest mechanism used for private market households. The total population identity $\sum_\tau (f_\tau^{s0} + f_\tau^{s1} + f_\tau^{L1}) = 1$ is preserved at convergence, where the convergence criterion of the stationary distribution is augmented to include the $L1$ layer with the same tolerance as the baseline.

In aggregation, $L1$ households count as homeowners in all statistics. Since $s = h$ is forced, they never appear as landlords. The government budget incorporates the RTB fiscal cost $Cost_{RTB} = \sum d \cdot p \cdot \tilde{h}'_{RTB} \cdot \pi_{RTB} \cdot f^{s1}$ over states with $\mathbb{I}_{RTB} = 1$, and adjusts RETT revenue to reflect that RTB buyers pay tax on the discounted price $p(1 - d)\tilde{h}'$ rather than on the market price. All RTB code is guarded by a conditional on the discount parameter d , ensuring that the baseline algorithm is recovered exactly when $d = 0$.

Equilibrium Adjustment. In the counterfactual experiments, policy changes alter household decisions, which in turn change housing demand and fiscal flows. With k_0 kept at its benchmark value, the construction technology implies a supply price $p_1 = k_0(I + I^s)^\phi$ that generally differs from the assumed p ; similarly, with the original tax parameters, the government budget G generally differs from the benchmark public good expenditure G_0 . Each counterfactual is solved under four equilibrium concepts: partial equilibrium (both p and λ_τ fixed at benchmark values), price adjustment only (p adjusts, λ_τ fixed), tax adjustment only (λ_τ scales, p fixed), and full general equilibrium (both adjust). In the specifications where adjustment is required, equilibrium is achieved by manual bisection over p and a uniform scaling factor applied to λ_τ , following the same approach as Kaas et al. (2021). Convergence is assessed on the scaled residuals $100 \times (p_1 - p)$ and $100 \times (G - G_0)$, both of which are required to fall below 0.05 in absolute value. This corresponds to a price discrepancy of less than 0.05% and a budget imbalance of less than 0.5 euros per household. When both p and λ_τ must adjust simultaneously, the two conditions are iterated in alternation: one variable is held fixed while the other is bisected toward its target, then the roles are reversed, exploiting the fact that the house price primarily governs housing demand while the tax scale primarily governs revenue.

I AI Usage

In the spirit of transparency, I adhere to a policy of full disclosure regarding the use of Artificial Intelligence (Model Gemini 3.1 Pro and Claude Opus 4.7) in the preparation of this proposal. I utilized AI tools strictly as auxiliary mechanisms to enhance technical efficiency and grammatical correctness, without replacing critical reasoning. Example prompts are listed below.

Code Debugging for Model solution

Please help identify the problem in the following lines of code:

[Area where specific code lines and the corresponding error message was provided.]

This prompt was used to debug code segments that did not initially work as intended.

LaTeX Formatting

Please typeset the following equation/line in LaTeX:

[Area where the text or equation was provided.]

There is an error in the LaTeX equation. Please help to solve the error.

Please transform this result *[screenshot of the MatLab output]* into a shorter and cleaner table.

How to link code-generated plots into LaTeX?

The tables take up substantial space in this paper, and I am constrained by a page limit. How to arrange these plots so as to adjust their size? *[Area where existing code lines for plotting were provided.]*

Please typeset the following citation in LaTeX format for the bibliography section of the paper: *[Area where the citation was provided.]*

Grammar Check

Please check this sentence for grammatical errors and redundancies.

[Area where the text was placed. Short excerpts (usually no more than 2–3 sentences at a time) from selected parts of the paper were provided.]

I confirm that the intellectual core of this proposal, including the research question, model extension and result interpretation is entirely human-generated. I guarantee that no part of the dataset was synthetically created or augmented by AI.

Replication Package

The quantitative framework of this thesis is largely based on the work of Kaas et al. (2021). Their replication package can be found under the following link:

<https://academic.oup.com/jeea/article/19/1/128/5821403?login=true#supplementary-data>

To access the code used to solve the model, produce the data, tables and figures in this thesis, please access the following Sciebo link, protected by the password **ypT38pV[&BC9**:

<https://uni-bonn.sciebo.de/s/s9oTdQ5j475Dqg3>

More information on how the code is structured can be found in the `README.pdf` files of the two main subfolders:

- `SHIW_Data_Creation`: includes the data from the SHIW and STATA file necessary to obtain homeownership profiles of Italy across age and wealth.
- `Matlab_Codes`: includes all the programs necessary to solve the model and produce tables and figures.

Statement of Authorship

I hereby confirm that the work presented has been performed and interpreted solely by myself except for where I explicitly identified the contrary. I assure that this work has not been presented in any other form for the fulfillment of any other degree or qualification. Ideas taken from other works in letter and in spirit are identified in every single case.

Place,	Date	Federico Proverbio
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